

LQR Learning Pipelines

between reinforcement learning & adaptive control

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papers

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PERSPECTIVES ON DATA-DRIVEN CONTROL

Scientific landscape

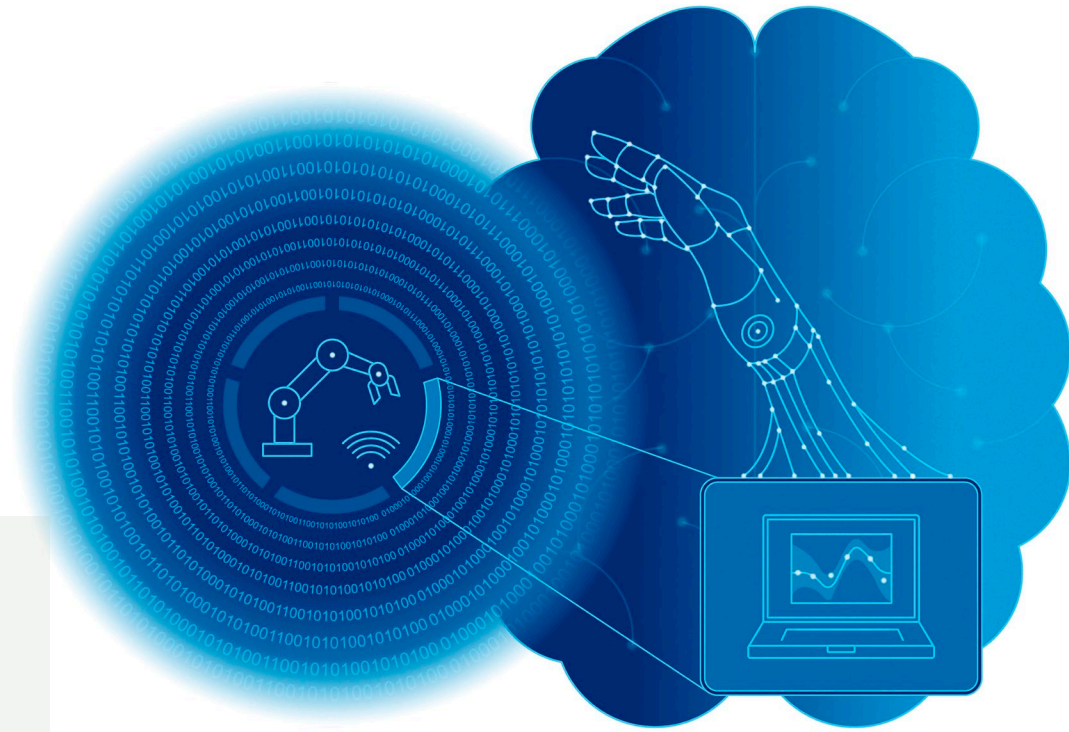
Rich & vast history

fragmented between fields: adaptive control & reinforcement learning (RL)

Culture gaps: adaptive control $\leftrightarrow$ RL

- stability vs optimality
- robust vs optimism facing uncertainty
- interpretable pen+paper vs compute
- theory certificates vs empirical studies

- **common root:** dynamic programming
- **early cross-overs:** neuro/adaptive DP



**Reinforcement Learning is Direct
Adaptive Optimal Control**

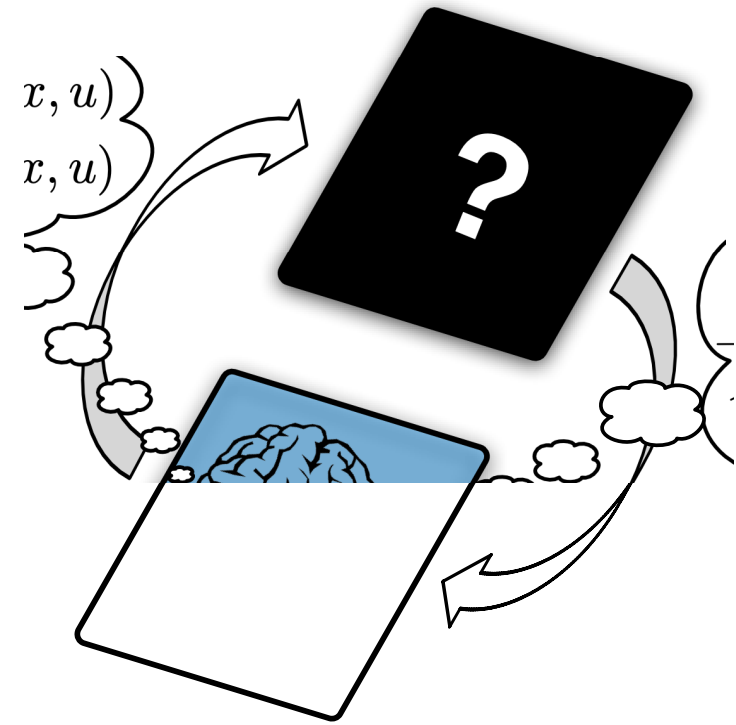
Richard S. Sutton, Andrew G. Barto, and Ronald J. Williams

- **today:** cross-fertilization & bridging culture gaps 4

Data-driven pipelines

- **indirect** (model-based) approach:
data $\xrightarrow{\text{ID}}$ model + uncertainty $\rightarrow$ control
- **direct** (model-free) approach:
direct MRAC, behavioral, RL...
- **episodic & batch** algorithms:
collect data batch $\rightarrow$ design policy

- **online & adaptive** algorithms:
measure $\rightarrow$ update policy $\rightarrow$ act

well-documented **trade-offs**

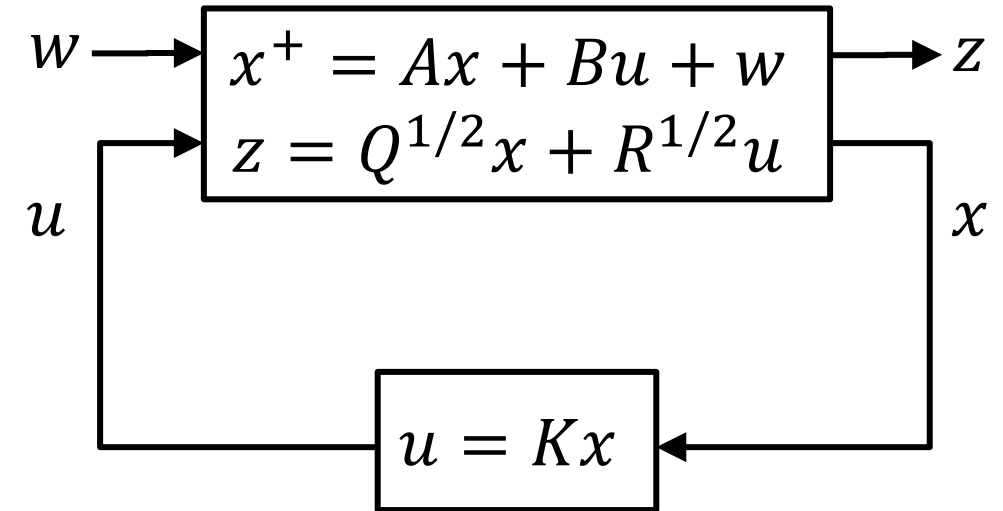
- goal: optimality vs robust stability
- practicality: modular vs end-to-end
- complexity: data, compute, theory

$\rightarrow$ **gold standard**: adaptive + optimal
+ robust + cheap + tractable ...& direct

Back to basics: LQR

- **cornerstone & the benchmark** of both optimal + adaptive control & RL

$$\begin{aligned} \underset{K}{\text{minimize}} \quad & \mathbb{E}_{w_t} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T x_t^T Q x_t + u_t^T R u_t \\ \text{s.t.} \quad & x^+ = Ax + Bu + w \\ & u = Kx \end{aligned}$$



- H_2 **parameterization** of LQR

direct
offline batch online adaptive
indirect

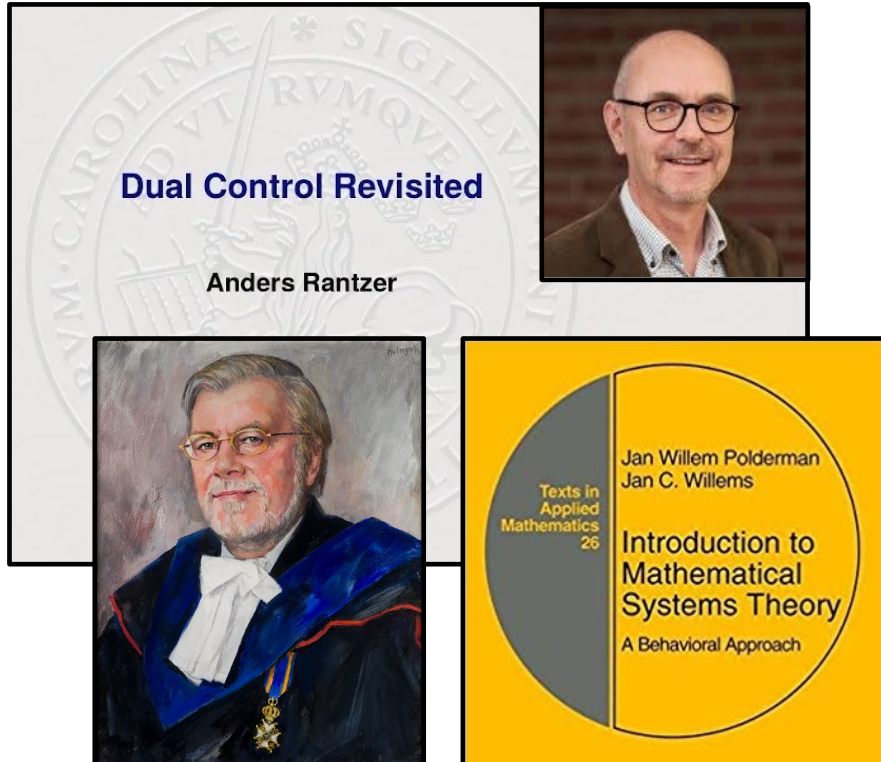
- **research gaps:** no direct + adaptive LQR & no closed-loop certificates of adaptive DP methods

Today: revisit old problems with new perspectives

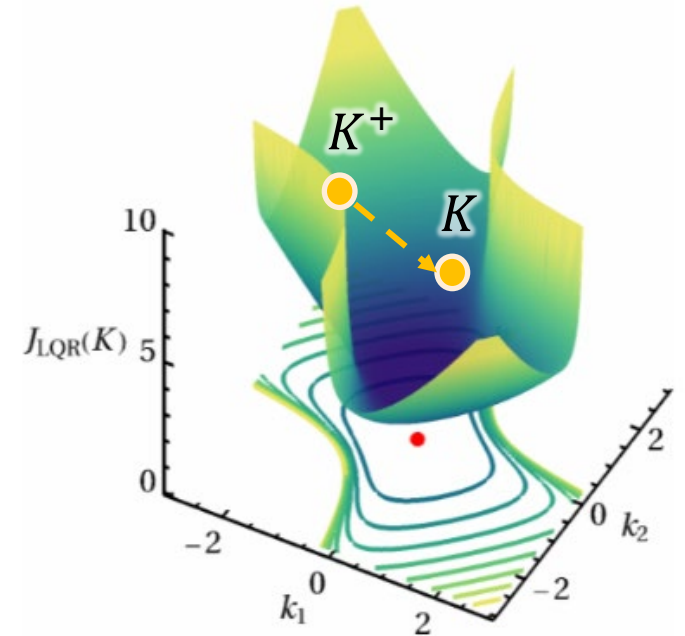
① **Behavior**: a linear system is a subspace of trajectories that can be represented by models or data: trajectory matrices or **sample covariances**

② **adaptation** of optimal control following RL-style **policy gradient descent**

$$K^+ = K - \eta \nabla J(K)$$



$\mathcal{X}$



[Watanabe & Zheng, 25]

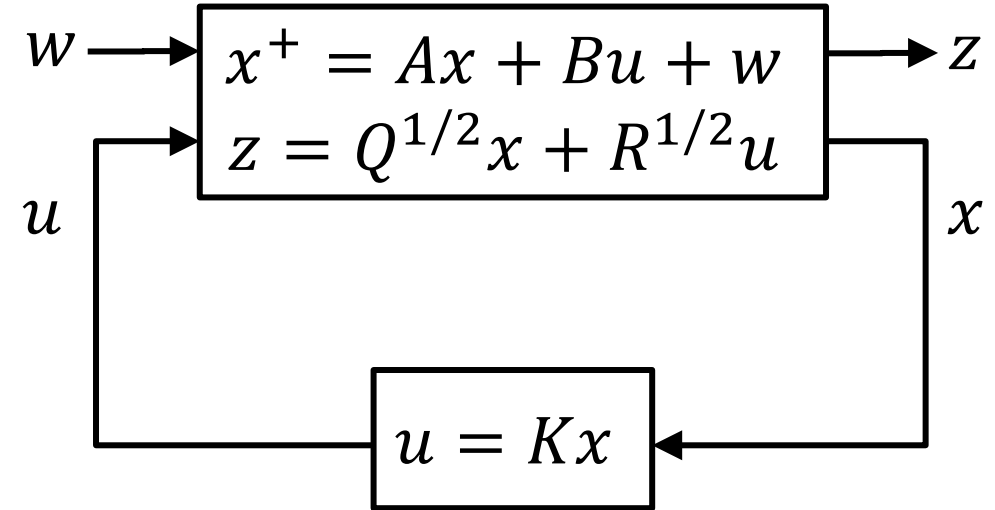
Contents

- 1. problem setup** for adaptive LQR via policy gradient
- 2. learning pipelines** for (in)direct policy gradients
- 3. closed loop:** sequential stability, optimality, & robustness
- 4. case studies:** numerics, robotics, flight, & power systems

PROBLEM SETUP FOR ADAPTIVE LQR VIA POLICY GRADIENT

LQR parameterization

$$\begin{aligned} \min_K \quad & \mathbb{E}_{w_t} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T x_t^T Q x_t + u_t^T R u_t \\ \text{s. t.} \quad & x^+ = Ax + Bu + w \\ & u = Kx \end{aligned}$$



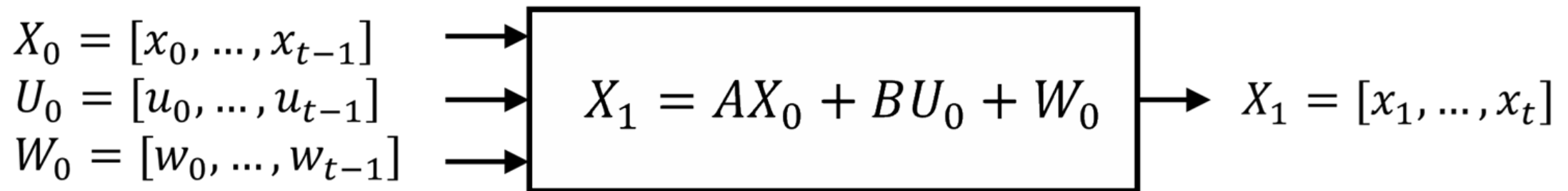
→ with **controllability Gramian** or **state covariance** $\Sigma = \lim_{T \rightarrow \infty} \frac{\sum_{t=0}^T x_t x_t^T}{T}$

$$\begin{aligned} \min_{K, \Sigma \succ 0} \quad & \text{Tr}(Q\Sigma) + \text{Tr}(K^T R K \Sigma) \\ \text{s. t.} \quad & \Sigma = I + (A + BK)\Sigma(A + BK)^T \end{aligned}$$

→ **model-based solution:** SDP or algebraic Riccati equation solved by interior point, contraction, or policy evaluation/improvement

Indirect & certainty-equivalence LQR

- **collect data** (X_0, U_0, X_1) with W_0 unknown & PE: $\text{rank} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} = n + m$



- **indirect & certainty-equivalence LQR**
(all solved offline)

$$\min_{K, \Sigma \succ 0} \text{Tr}(Q\Sigma) + \text{Tr}(K^T R K \Sigma)$$

$$\text{s. t. } \Sigma = I + (\hat{A} + \hat{B}K)\Sigma(\hat{A} + \hat{B}K)^T$$

$$[\hat{B}, \hat{A}] = \arg \min_{B, A} \left\| X_1 - [B, A] \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} \right\|_F$$

certainty-
equivalent
LQR

least
squares
SysID

Towards an online & adaptive implementation

- **shortcoming** of separating learning and control
→ cannot improve policy **online** & cheaply / rapidly **adapt**
- **monotonicity principles** of adaptive control: acquire information & improve control performance over time
- **desired adaptive** solution: online (non-episodic / batch) algorithm, with closed-loop data, recursive implementation, & (in?)direct
- how to “best” improve policy online → **go down the gradient !**

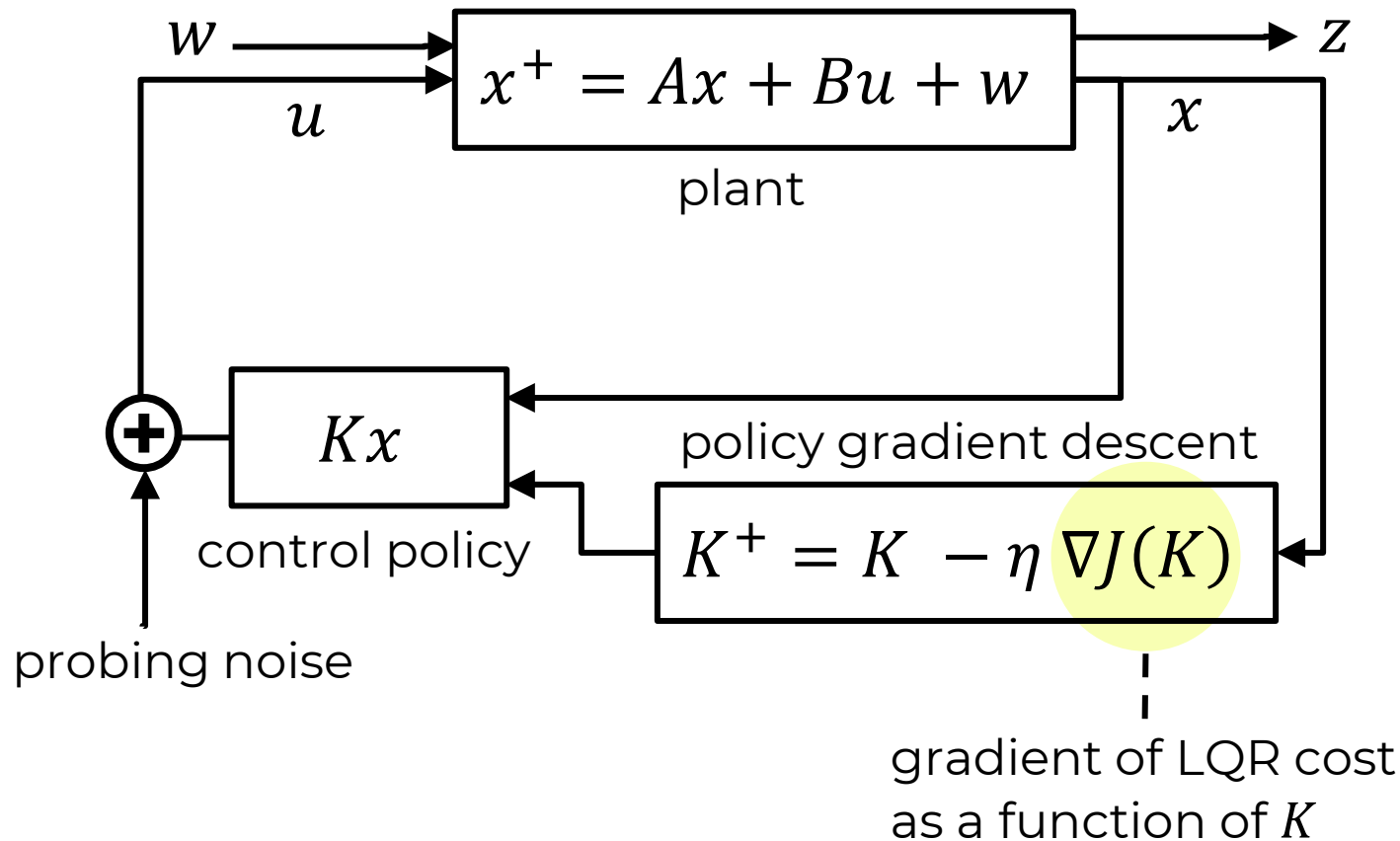


A daptive C ontrol:
T ow ards a C om plexity-B ased G eneral T heory*

G. ZAMES-

* disclaimer: a large part of the adaptive control community focuses on stability & not optimality

Adaptive LQR via policy gradient descent



Seems obvious but...

→ **algorithms**

- how to compute $\nabla J(K)$ cheaply & recursively?
- direct or indirect?
- convergence?

→ **closed loop**

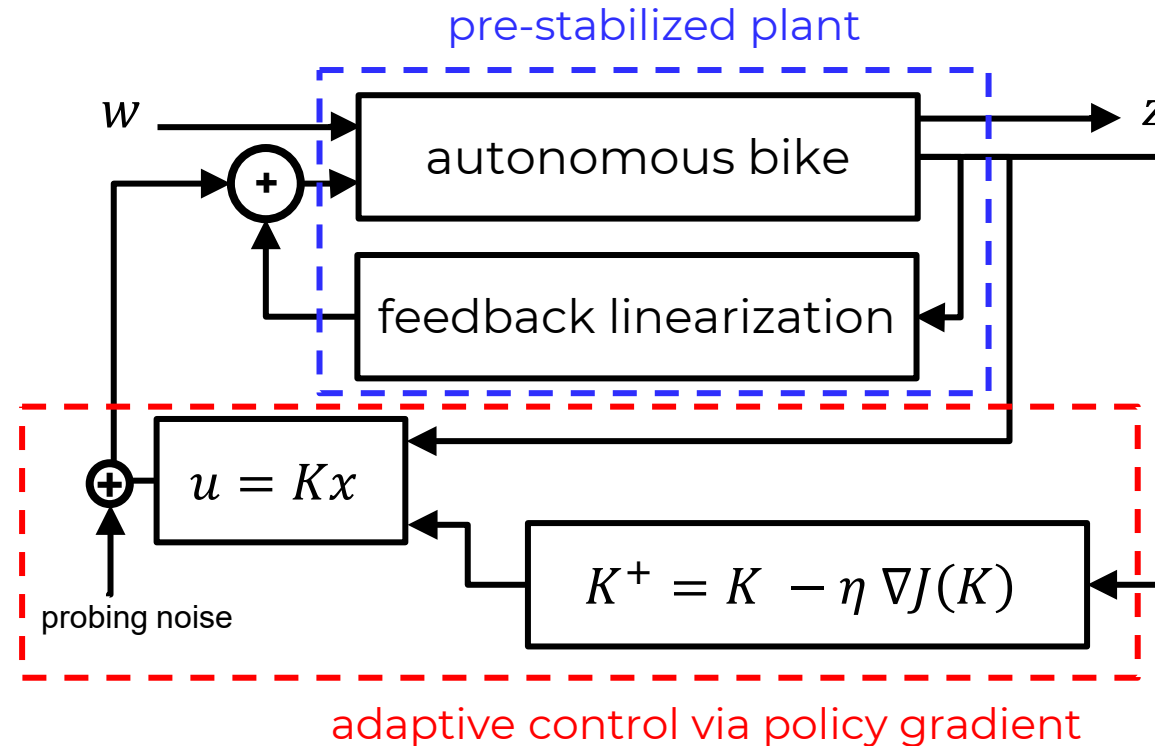
- stability?
- robustness?
- optimality?

Preview: does it work on an autonomous bike ?



Hardware

① RC receiver	⑤ Bafang RM G040.250.DC
② Raspberry Pi 4b	⑥ Xsens MTi-7
③ ESC	⑦ Batteries
④ Hall sensor	⑧ Dynamixel XH540-W270-T



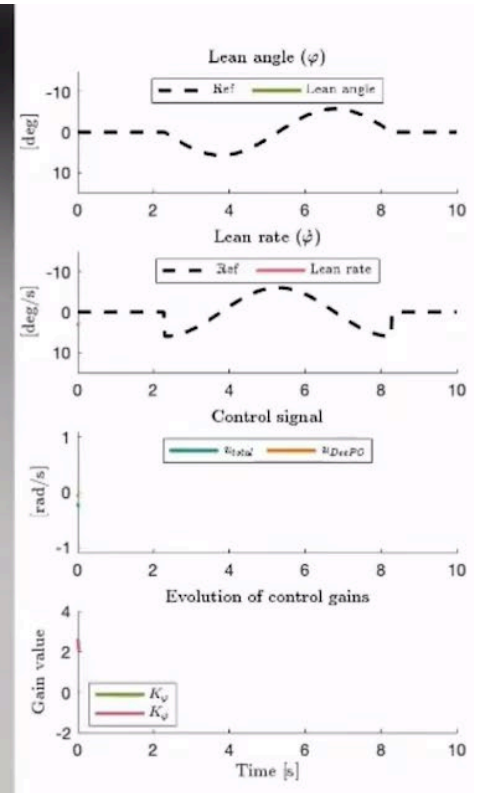
Setup: **autonomous bicycle** with **coarse inner control** (2d dynamics stabilized by feedback linearization) & **outer adaptive policy gradient**

Preview: does it work on an autonomous bike ?



Hardware

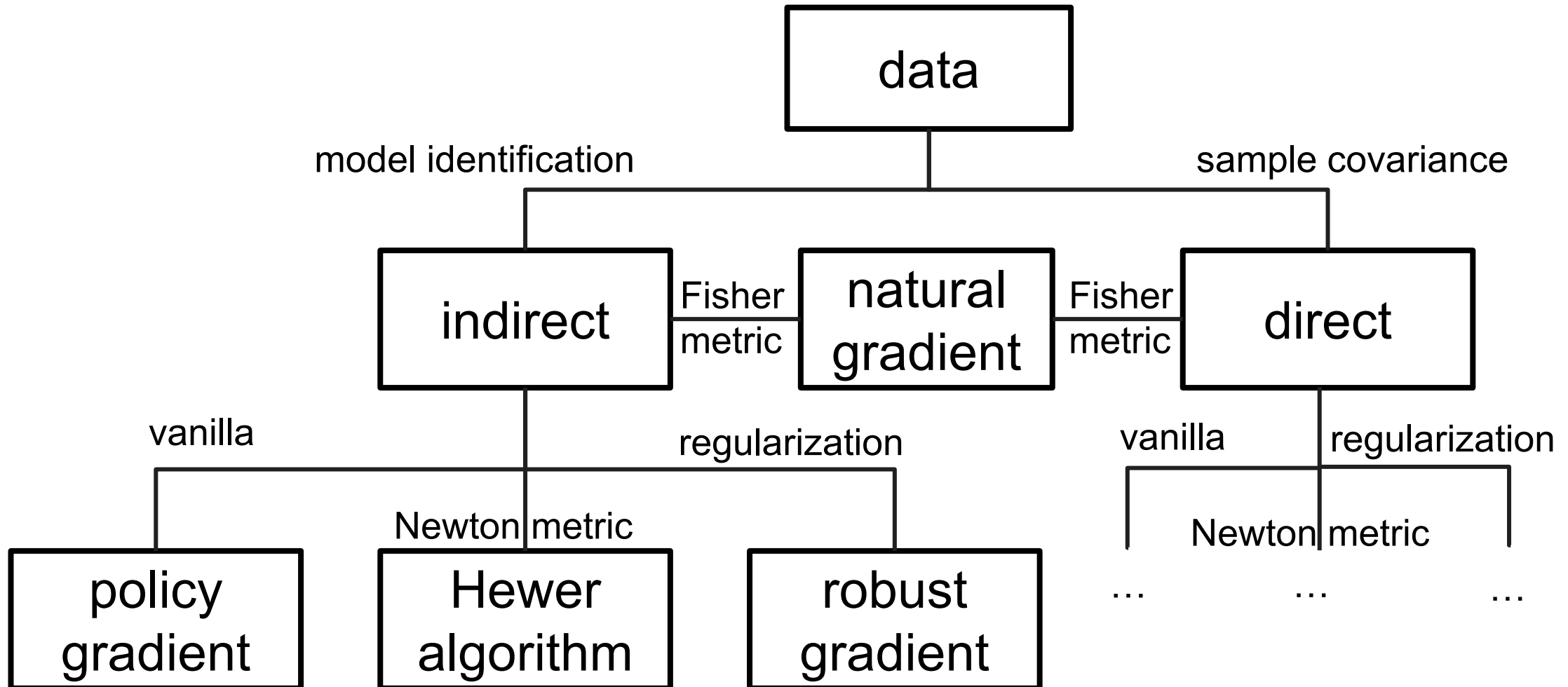
①	RC receiver	⑤	Bafang RM G040.250.DC
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③	ESC	⑦	Batteries
④	Hall sensor	⑧	Dynamixel XH540-W270-T



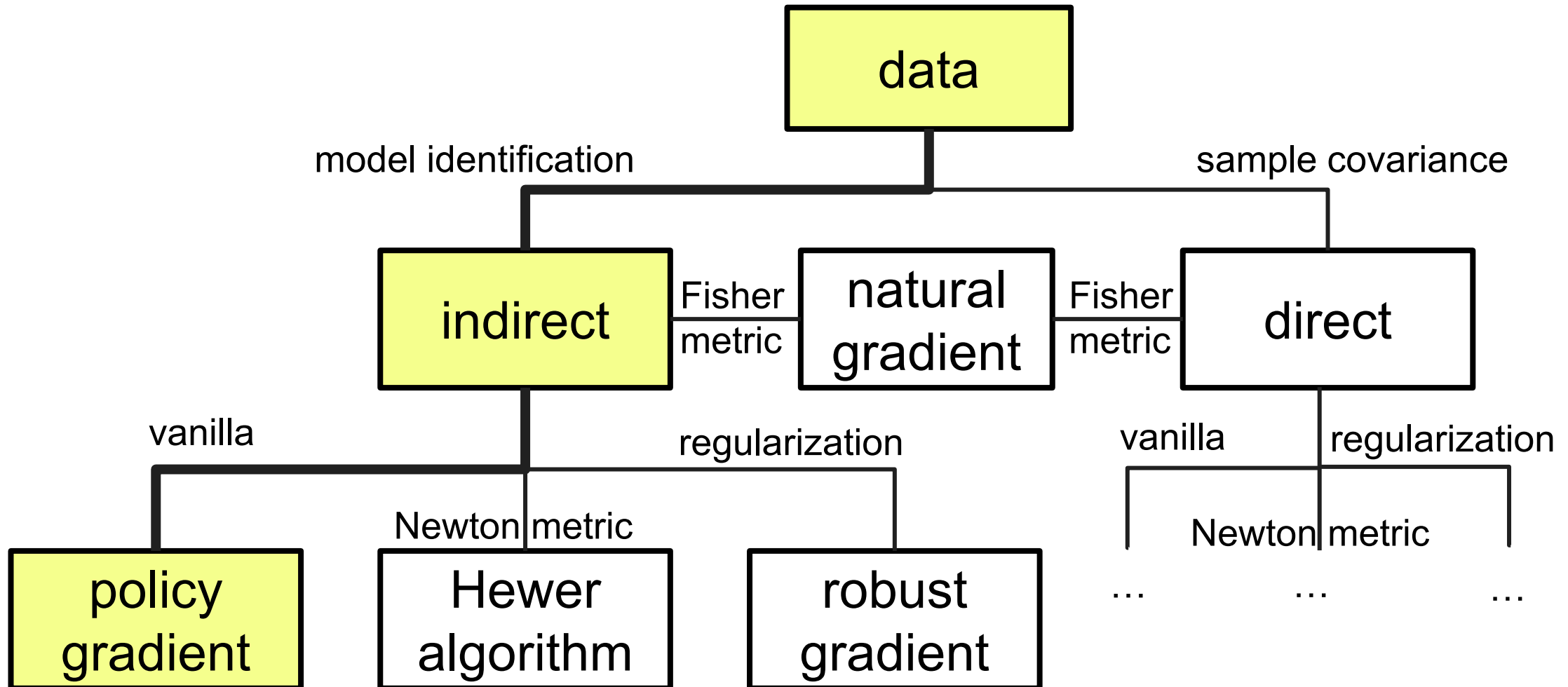
Setup: **autonomous bicycle** with **coarse inner control** (2d dynamics stabilized by feedback linearization) & **outer adaptive policy gradient**

LEARNING PIPELINES FOR POLICY GRADIENT

Algorithmic road map



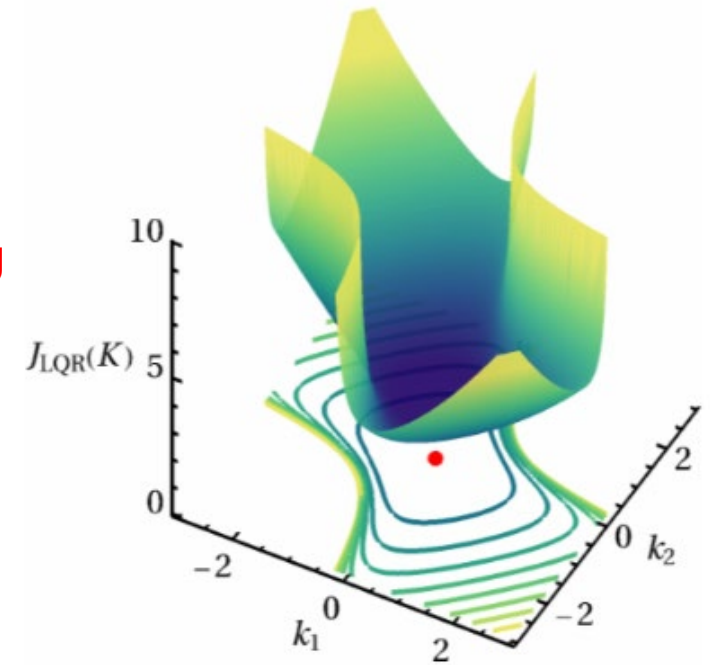
Algorithmic road map



LQR optimization landscape

$$\begin{aligned} \min_{K, \Sigma \succ 0} \quad & J(K) = \text{Tr}(Q\Sigma) + \text{Tr}(K^T R K \Sigma) \\ \text{s. t.} \quad & \Sigma = I + (A + BK)\Sigma(A + BK)^T \end{aligned}$$

after eliminating
unique $\Sigma \succ 0$,
denote the
objective as $J(K)$



$J(K)$ is usually not convex but for stabilizing K is

- **differentiable** with $\nabla J(K) = 2 \left((R + B^T P B) K - B^T P A \right) \Sigma$ [Watanabe & Zheng, 25]

where $P = Q + K^T R K + (A + BK)^T P (A + BK)$

& Σ are closed-loop obs. + contr. Gramians

- **coercive** with compact sublevel sets
- smooth with locally **bounded Hessian**
- **gradient dominance**: $J(K) \leq J^* + \text{const.} \cdot \|\nabla J(K)\|^2$

Gradient methods for iterative distributed control synthesis

Karl Mårtensson and Anders Rantzer

Model-based policy gradient

$$\min_K J(K) = \text{Tr}(Q\Sigma) + \text{Tr}(K^T R K \Sigma)$$

where $\Sigma = I + (A + BK)\Sigma(A + BK)^T$

Fact: For initial K_0 stabilizing & small η , **gradient descent** $K^+ = K - \eta \nabla J(K)$ **converges linearly** to K^* .

Annual Review of Control, Robotics and Autonomous Systems

Toward a Theoretical
Foundation of Policy
Optimization for Learning
Control Policies

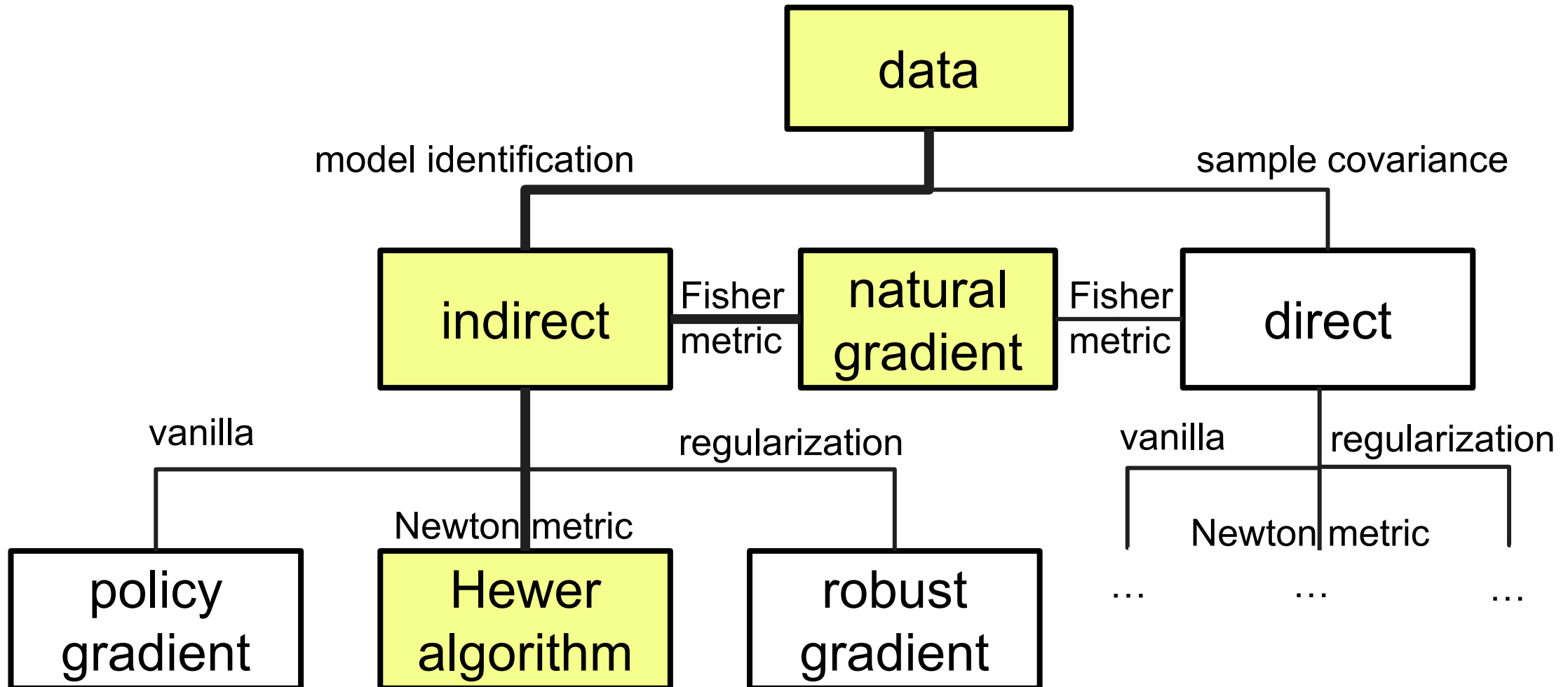
Bin Hu,¹ Kaiqing Zhang,^{2,3} Nali,⁴ Mehran Mesbahi,⁵
M. Aryan Fazel,⁶ and Tamer Başar¹

Algorithm: model-based adaptive control via policy gradient

1. **data collection:** refresh (X_0, U_0, X_1)
2. **identification** of $(\hat{B}, \hat{A})$ via recursive LS
3. **policy gradient:** $K^+ = K - \eta \nabla J(K)$ using estimates $(\hat{B}, \hat{A})$ & closed-loop Gramians (Σ, P)

actuate
& repeat

Algorithmic road map



Pre-scaled policy gradient

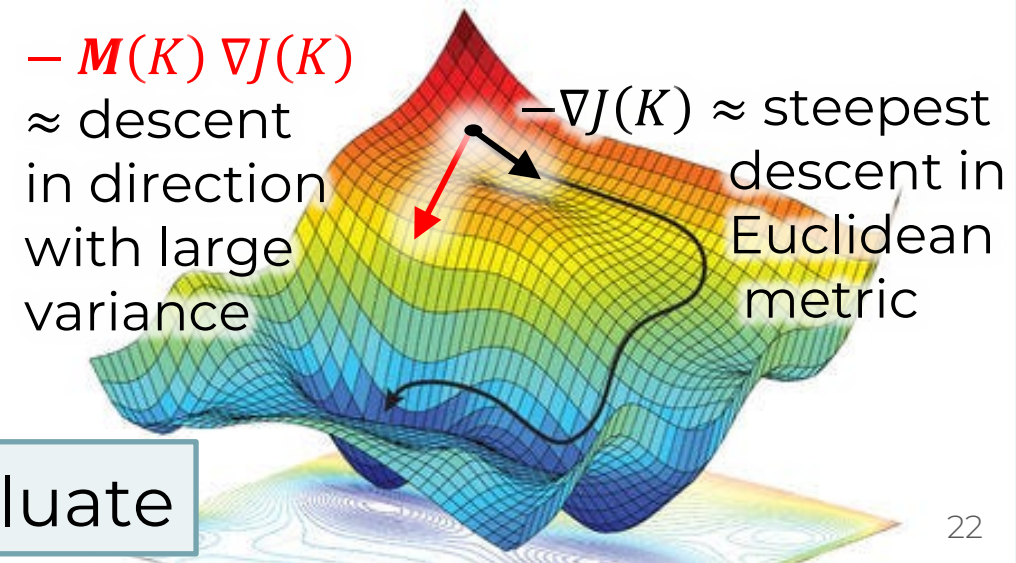
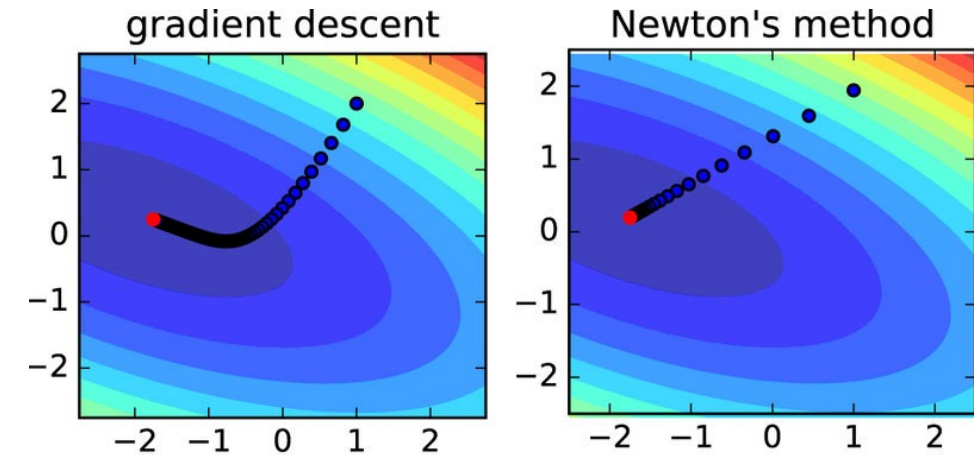
$$K^+ = K - \eta \mathbf{M}(K) \nabla J(K)$$

→ **Gauss-Newton**: $\mathbf{M}(K)$ cheaply approximates inverse Hessian $\nabla^2 J(K)$

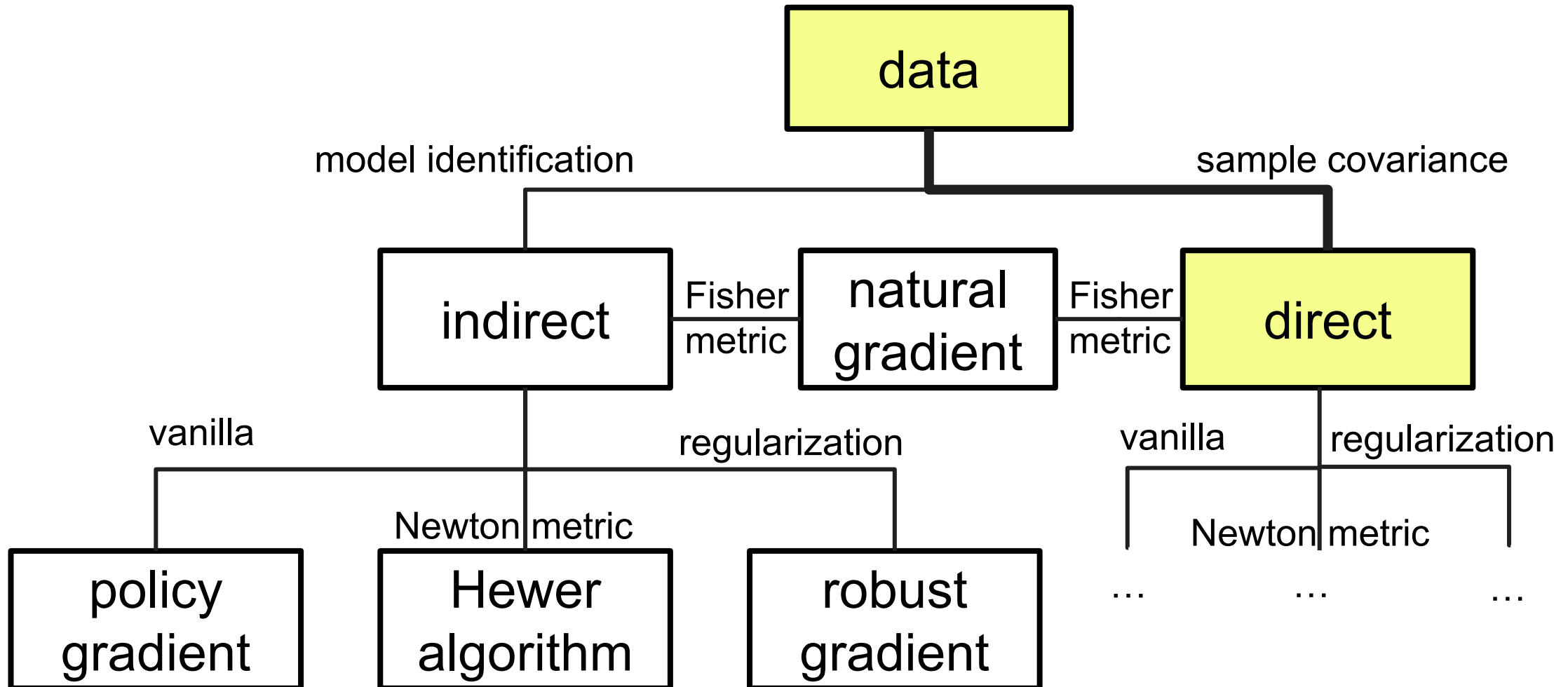
Fact: this equals Hewer's algorithm:
policy evaluation $\Leftrightarrow$ improvement

→ **Natural policy gradient** method:
 $\mathbf{M}(K)$ is inverse Fisher information

Fact: $\mathbf{M}(K) \nabla J(K) = \nabla J(K) \Sigma^{-1}$ is easy to evaluate



Algorithmic road map



Direct (model-free) policy gradient methods

- **issue:** uncertainty propagation is hard in indirect case & outcome suffers from bias error (e.g. model-order in output feedback setup)

- **model-free 0th order methods** constructing two-point estimate

$$\widehat{\nabla J(K)} = (J(K + rU) - J(K - rU)) \cdot \frac{mn}{r^2} U$$

from uniform perturbation U & numerous + very long trajectories

relative performance gap	$\epsilon = 1$	$\epsilon = 0.1$	$\epsilon = 0.01$
# trajectories (100 samples)	1414	43850	142865

**$\sim 10^7$ samples for
4th order system**

- direct policy gradient is inefficient, episodic, & **practically useless ?**
→ **sample covariance parameterization to the rescue !**

Sample covariance parametrization of the LQR_(no noise)

$$\begin{array}{l} X_0 = [x_0, \dots, x_{t-1}] \\ U_0 = [u_0, \dots, u_{t-1}] \end{array} \longrightarrow \boxed{X_1 = AX_0 + B U_0} \longrightarrow X_1 = [x_1, \dots, x_t]$$

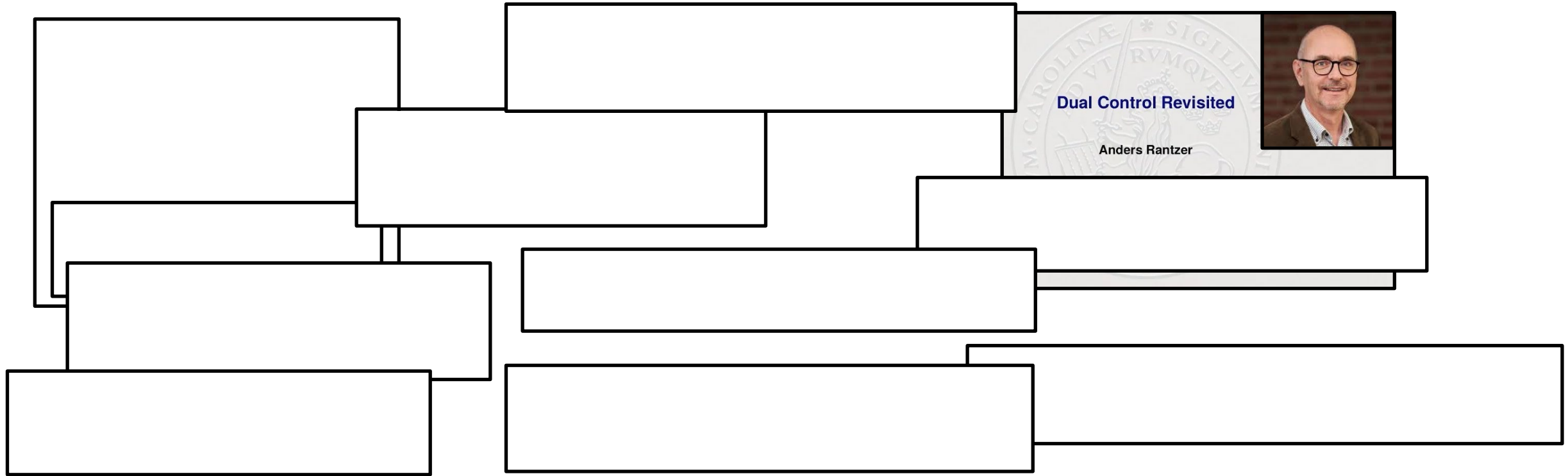
- sample covariances:** $\Phi = \frac{1}{t} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\top \succ 0$ & $\Phi^+ = \frac{1}{t} X_1 \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\top$
(due to PE)

$$\Rightarrow \forall K \exists V \text{ s.t. } \begin{bmatrix} K \\ I \end{bmatrix} = \frac{1}{t} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\top V \quad (\star)$$

- closed loop:** $A + BK = [B \ A] \begin{bmatrix} K \\ I \end{bmatrix} \stackrel{\text{green box}}{=} [B \ A] \frac{1}{t} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\top V \stackrel{\text{red box}}{=} \frac{1}{t} X_1 \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\top V = \Phi^+ V$

- direct data-driven** formulation by substituting $A + BK = \Phi^+ V$ & $(\star)$

Aside: covariance parametrization on the rise



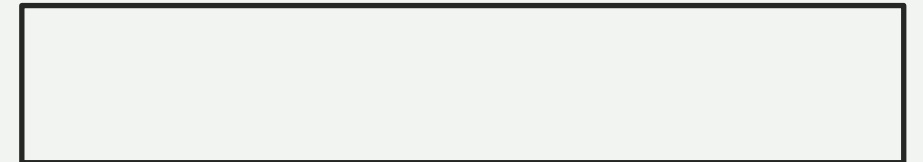
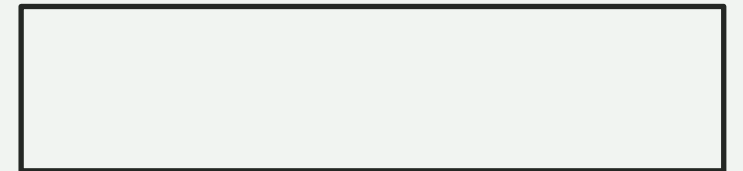
Dual Control Revisited

Anders Rantzer



Related to trajectory matrices, but

- matrices independent of data size
- uniqueness & no regularization
- recursive rank-1 updates



Covariance parametrization of policy gradient

- covariance parameterization:**

substitute $A + BK = \Phi^+ V$ with

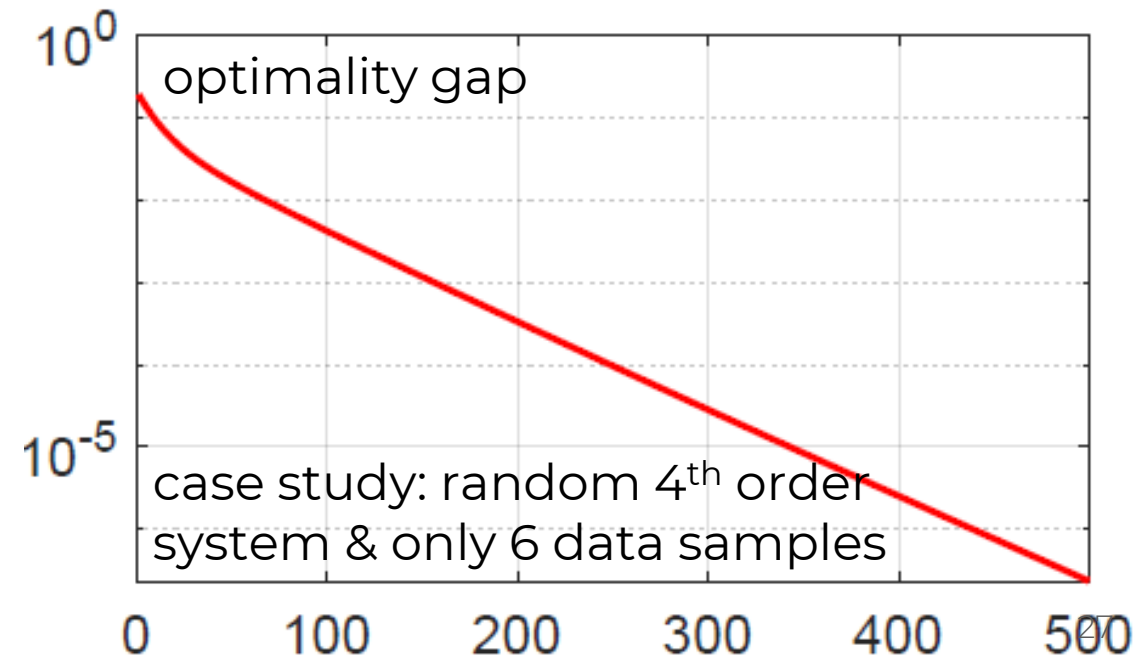
linear constraint $\begin{bmatrix} K \\ I \end{bmatrix} = \Phi V$

- analogous optimization problem in V -coordinates with linear constraint, where K can be eliminated

- direct & projected policy gradient**

$$V^+ = V - \eta \Pi(\nabla J(V))$$

$$\begin{aligned} & \min_{K, \Sigma \succ 0} \text{Tr}(Q\Sigma) + \text{Tr}(K^T R K \Sigma) \\ & \min_{K, \Sigma \succ 0} \text{Tr}(Q\Sigma) + \text{Tr}(K^T R K \Sigma) \\ & \text{s. t. } \Sigma = I + (\Phi^+ V) \Sigma (\Phi^+ V)^T \\ & \text{s. t. } \Sigma = I + \begin{bmatrix} K \\ I \end{bmatrix} \begin{bmatrix} K^T & I^T \end{bmatrix} \Sigma \begin{bmatrix} K \\ I \end{bmatrix} \end{aligned}$$



Facts on direct policy gradient

- **covariance parameterization:** substitute $A + BK = \Phi^+ V$ with $\begin{bmatrix} K \\ I \end{bmatrix} = \Phi V$
- **direct & projected policy gradient descent** $V^+ = V - \eta \Pi (\nabla J(V))$

Fact 1: in original K –coordinates reads as **scaled gradient descent**

$$K^+ = K - \eta \mathbf{M}_t \nabla J(K),$$

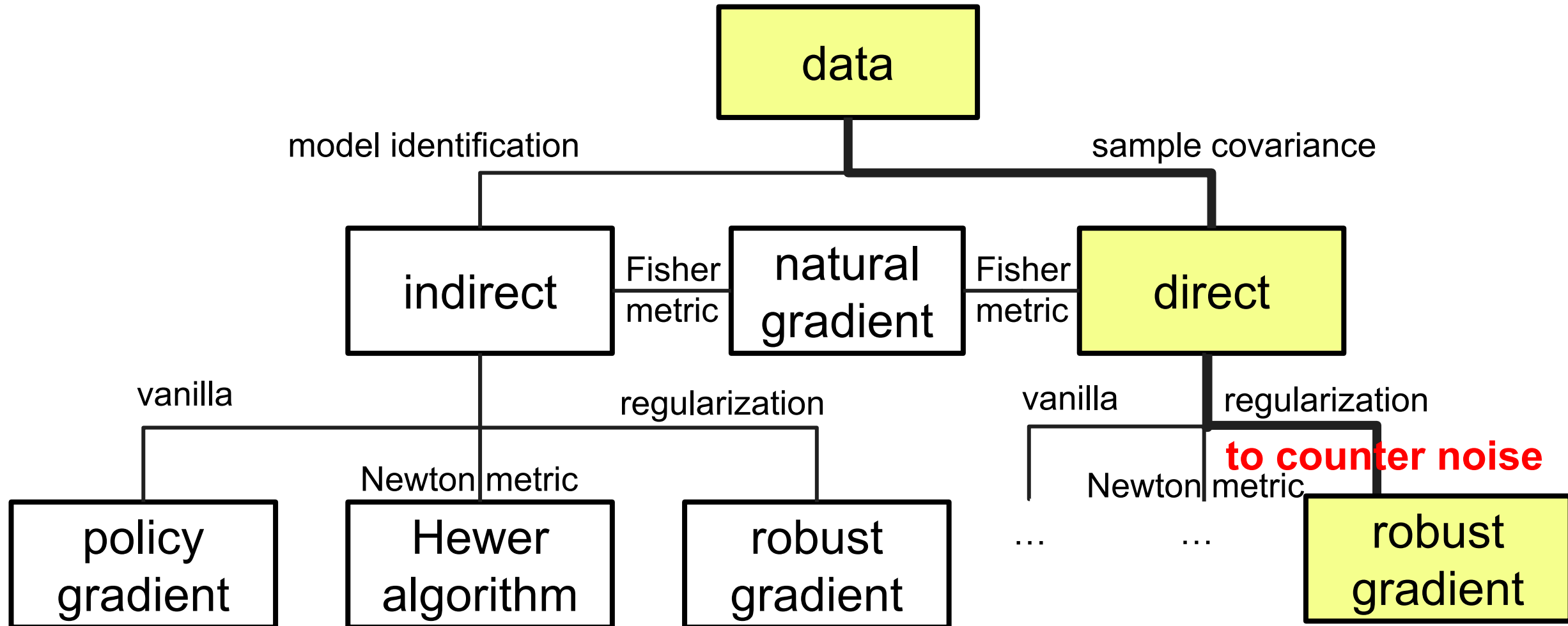
where $\mathbf{M}_t = \frac{1}{t^2} U_0 \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\top \Pi \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} U_0^T > 0$.

→ retain strong convergence result

Fact 2: the corresponding **natural gradient descent is invariant**, i.e., equal to the model-based natural gradient descent.

→ recover original result

Algorithmic road map



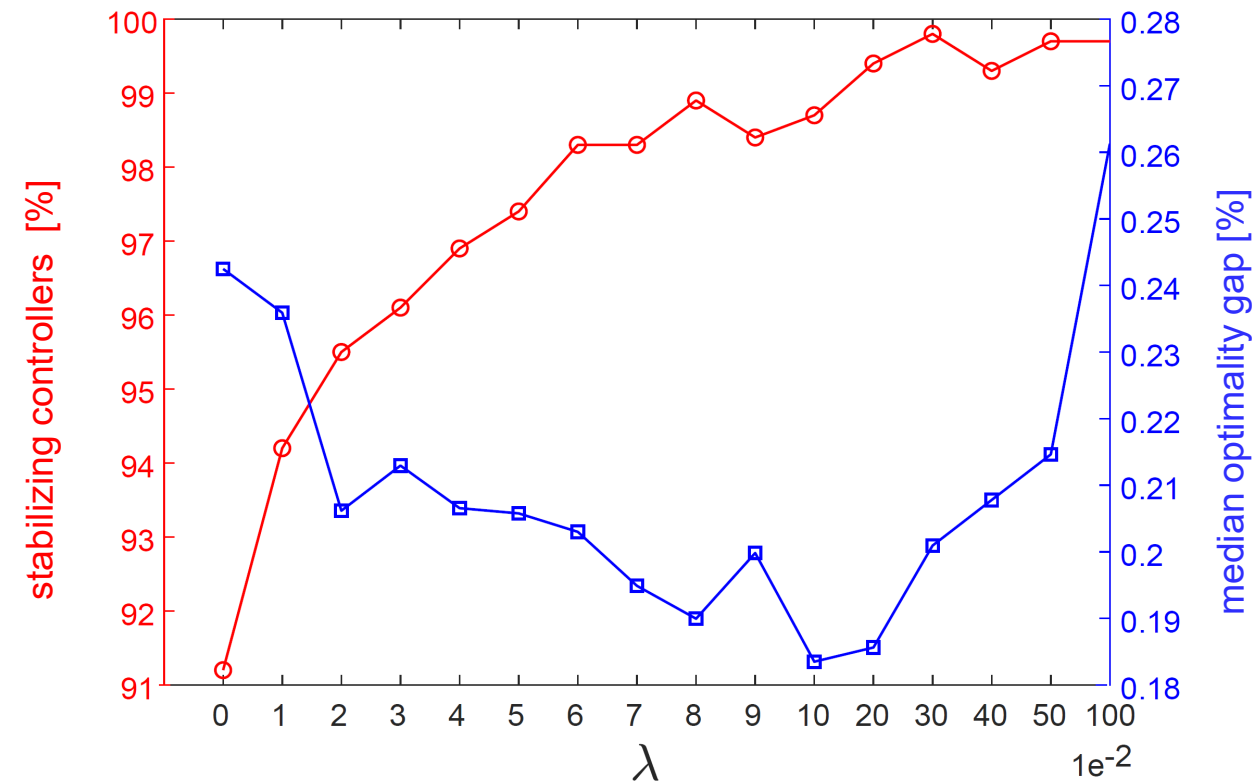
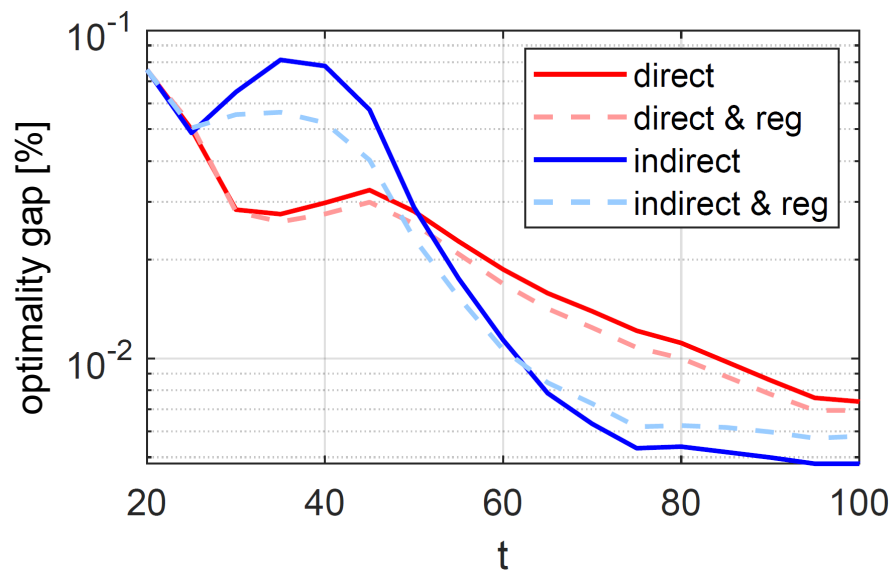
Robustifying covariance regularization of the LQR

$$\begin{array}{l} X_0 = [x_0, \dots, x_{t-1}] \longrightarrow \\ U_0 = [u_0, \dots, u_{t-1}] \longrightarrow \\ W_0 = [w_0, \dots, w_{t-1}] \longrightarrow \end{array} \boxed{X_1 = AX_0 + B U_0 + W_0} \longrightarrow X_1 = [x_1, \dots, x_t]$$

- **closed loop:** $A + BK = (\Phi^+ - \tilde{W}_0)V$ where $\tilde{W}_0 = \frac{1}{t} W_0 \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^T$ (neglected before)
- **difference in Lyapunov equn's** with/without noise $\sim V \Sigma V^T \Phi$
- **regularized LQR:** $J(V) + \lambda \cdot \text{Tr}(V \Sigma V^T \Phi) = J(K) + \lambda \cdot \text{Tr}\left(\Phi^{-1} \begin{bmatrix} K \\ I \end{bmatrix} \Sigma \begin{bmatrix} K \\ I \end{bmatrix}^T\right)$
- optimization class does not change $\rightarrow$ **retain convergence result**

Effect of regularization

- regularizing with λ robustifies & gives better performance
- improves algorithmic stability for indirect & direct methods
- decrease λ as data grows

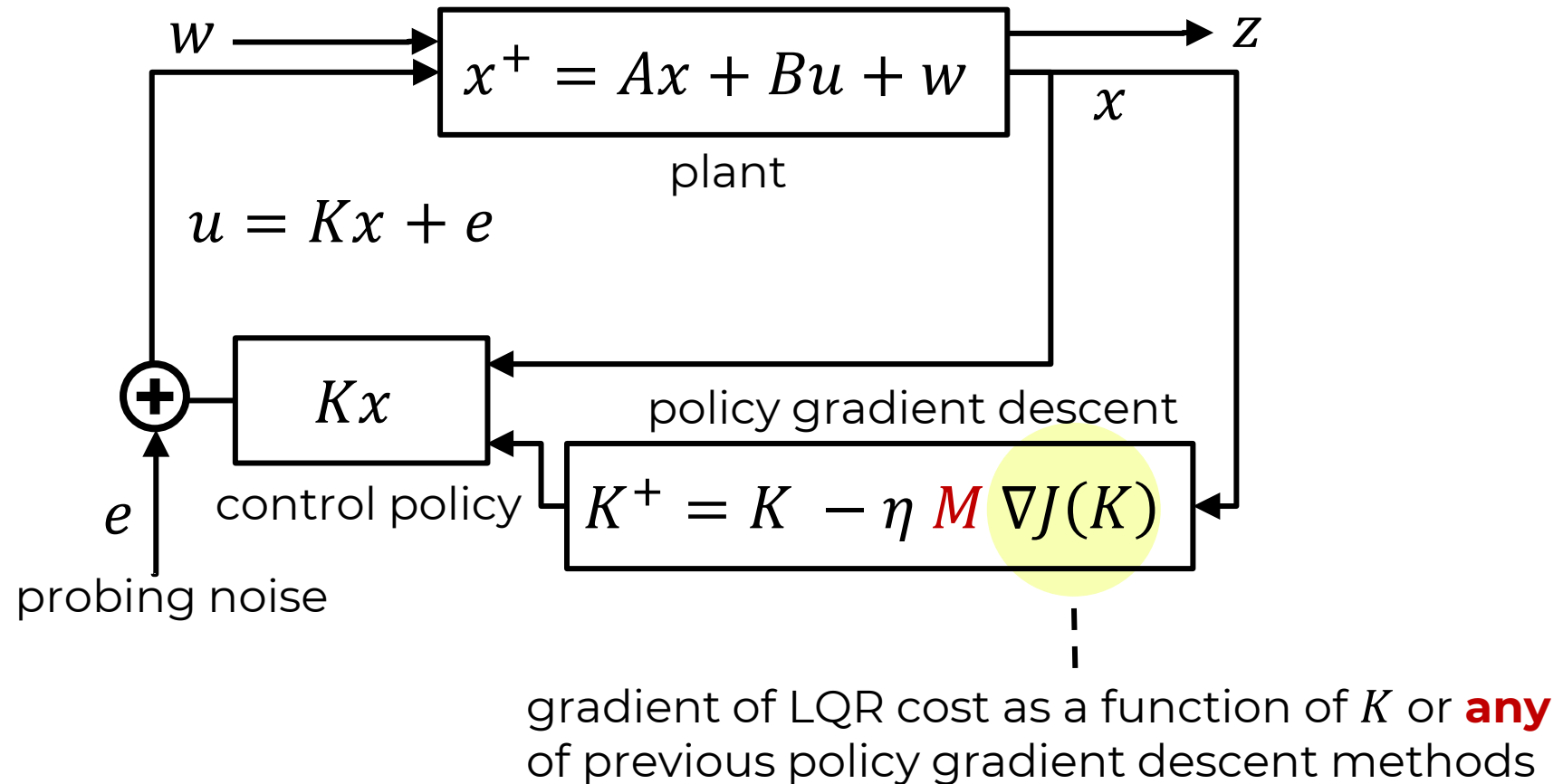


Comparison of stability % $\mathcal{P}$ & median optimality gap $\mathcal{M}$

	$\lambda_t = 0$	$\lambda_t = 1/(10\sqrt{t - t_0})$	one-shot-based
direct	$\mathcal{P} = 81\%$ $\mathcal{M} = 0.0014$	$\mathcal{P} = 98\%$ $\mathcal{M} = 0.0011$	$\mathcal{P} = 82\%$ $\mathcal{M} = 0.0013$
indirect	$\mathcal{P} = 83\%$ $\mathcal{M} = 0.0014$	$\mathcal{P} = 99\%$ $\mathcal{M} = 0.0014$	$\mathcal{P} = 82\%$ $\mathcal{M} = 0.0013$

CLOSED-LOOP STABILITY, OPTIMALITY & ROBUSTNESS

Policy gradient descent in closed loop



Q: if each K_t is stabilizing & $J(K_t)$ decreases, we surely get asymptotic stability & optimality ?

A: it's a switched system, ... so no ? 33

Information metric

- **persistency of excitation** due to probing:

$$\underline{\sigma}(\Phi_t) \geq \gamma_t \text{ for some } \gamma_t > 0$$

- **noise & data covariance:**

$$\delta_t := \left\| \frac{1}{t} W_0 D_0^T \right\|$$

- information metric = **signal-to-noise ratio** $SNR_t := \gamma_t / \delta_t$

satisfies Zames' first
monotonicity principle:
information acquisition
= SNR increases



A daptive Control:
Tow ards a C om plexity-Based G eneral Theory*

G. ZAMES-

Uniform $\delta_t \sim O(1)$	Constant $\gamma_t \sim O(1)$	$SNR_t \sim O(1)$
Gaussian $\delta_t \sim O(1/\sqrt{t})$	Constant $\gamma_t \sim O(1)$	$SNR_t \sim O(\sqrt{t})$
	Decay $\gamma_t \sim O(t^{-1/4})$	$SNR_t \sim O(t^{1/4})$

Certificate for **any** of the policy gradient methods

Theorem (simplified): There exist $v_i > 0, i \in \{1,2,3,4,5\}$, depending on (A, B, Q, R, K_0) with $v_4 < 1$, so that, if $SNR_t \geq v_1 \forall t$, $\eta \leq v_2$, & for stable K_0

1. the **closed-loop system is stable** in the sense that

$$|x_t| \leq v_3 v_4^t |x_0| + v_5 \max_{0 \leq i < t} |Be_i + w_i|.$$

SNR & step-size requirements

2. the **policy converges to optimality** in the sense that

$$C(K_t) - C^* \leq \left(1 - \frac{\eta}{v_6}\right)^t (C(K_{t_0}) - C^*) + O(SNR_t^{-1})$$

stable initialization

nominal exponential convergence

bias due to noise

Notes on stability & convergence statement

- **Key to the analysis:** small enough step size to control learning rate & assure **sequential stability**
- **convergence:** nominal exponential + (decreasing) bias term
 - Zames' 2nd monotonicity principle: improving performance
 - $O(1/\sqrt{t})$ for Gaussian noise & constant excitation
 - $O(t^{-1/4})$ for Gaussian noise & diminishing excitation

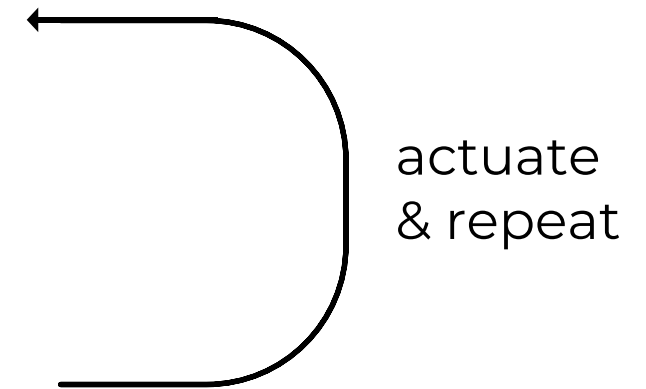
slightly worse
than optimal
known rates
 $O(1/t)$ & $O(1/\sqrt{t})$
- **direct methods:** $SNR_t \geq \frac{\text{const.} \|M_t\|}{\underline{\sigma}(M_t)}$, $\eta_t \leq \frac{\text{const.}}{\|M_t\|}$, & convergence rate
depend on data-dependent matrix $M_t = \frac{1}{t^2} U_0 \begin{bmatrix} U_0 \\ X_0 \end{bmatrix}^\top \Pi \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} U_0^T$

Notes on stability & convergence statement

- all results also hold in **regularized setting** under proper choice of regularization coefficient $\lambda_t \leq O(\delta_t)$ (const. for bounded noise)
- Indirect Gauss-Newton method = adaptive version of **Hewer's algorithm**, which needs additionally K_0 sufficiently close to K^*

Algorithm: adaptive Hewer's algorithm

1. **data collection:** refresh (X_0, U_0, X_1)
2. **identification** of $(\hat{B}, \hat{A})$ via recursive LS
3. **policy evaluation:** $P^+ = \text{Lyapunov}(\hat{B}, \hat{A}, K)$
4. **policy improvement:** $K^+ = (\dots)^{-1} P^+ \hat{B} \hat{A}$

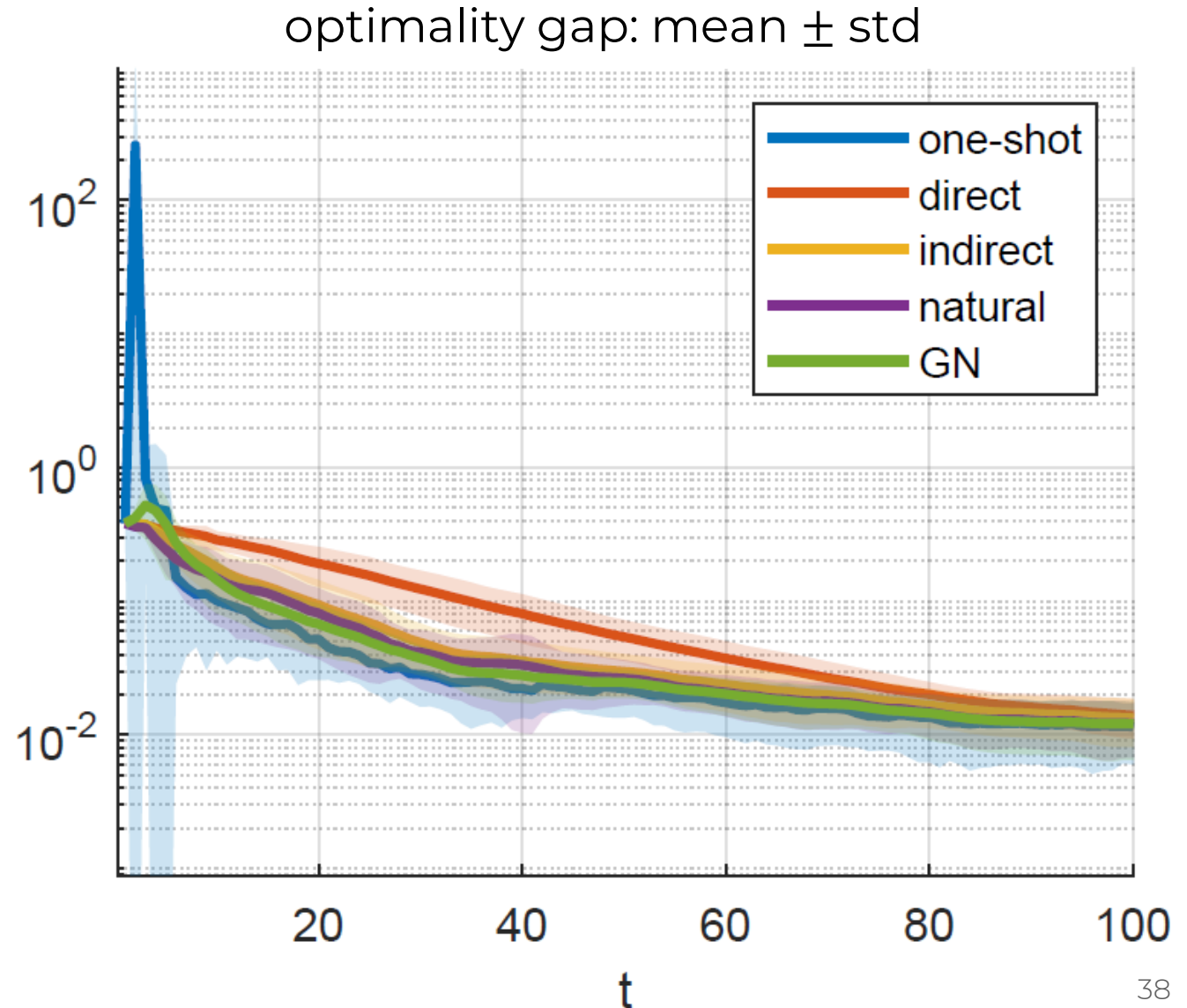


Numerics: convergence to optimality

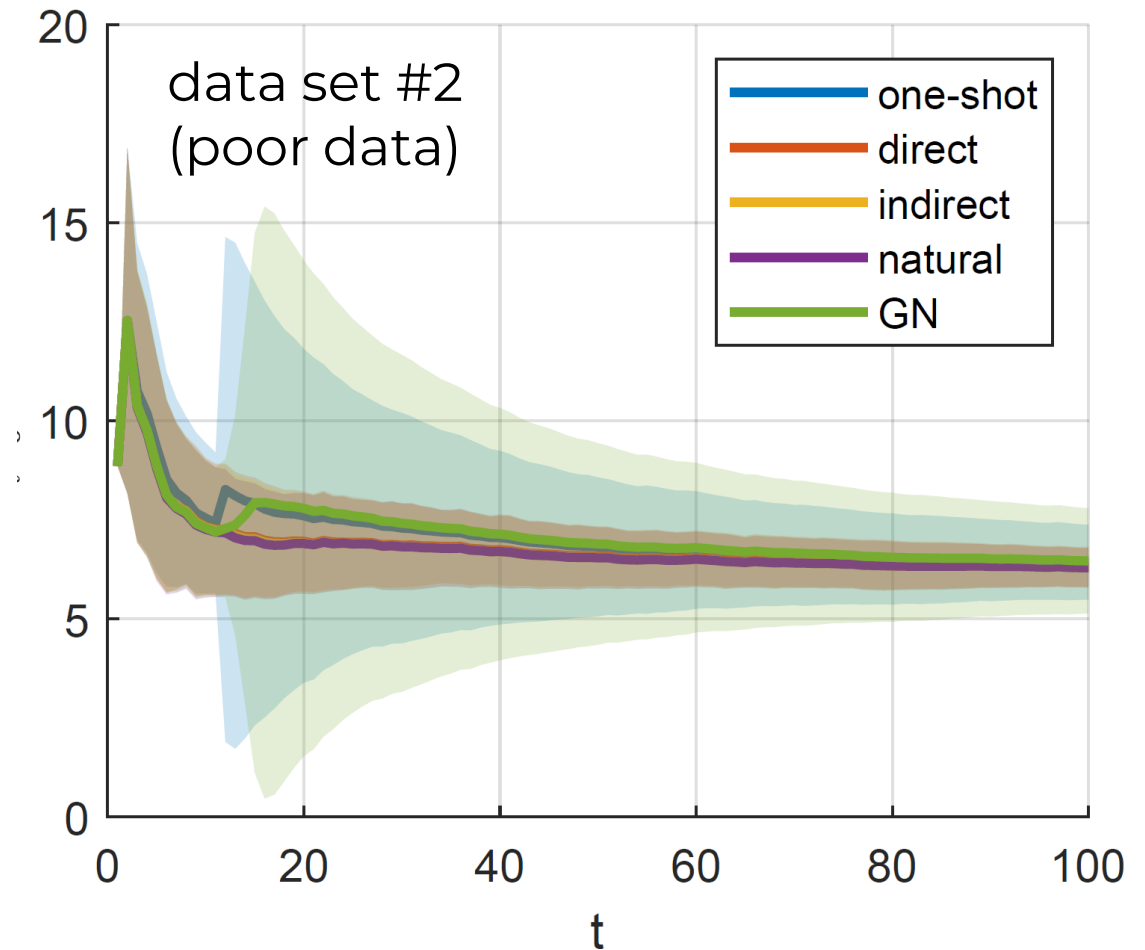
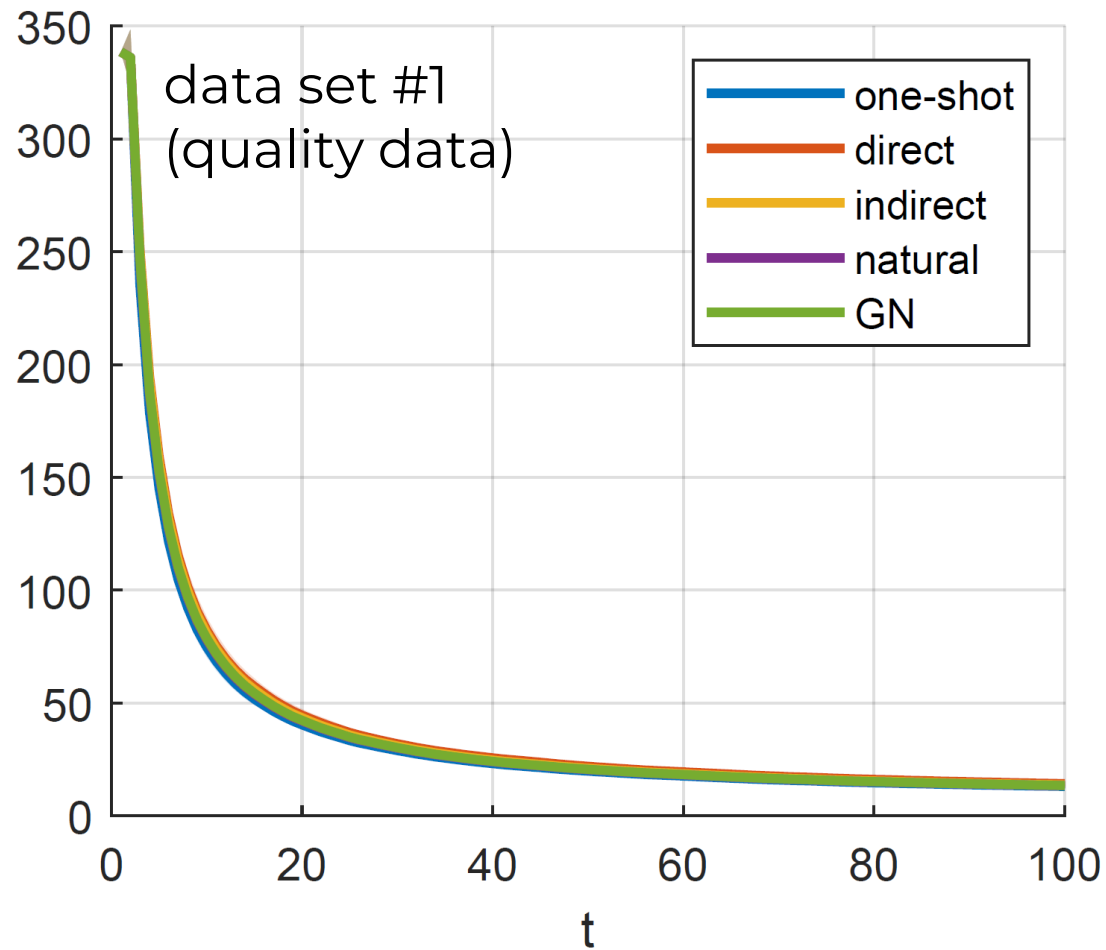
- **case study** [Dean et al. '19]:
discrete-time system with
Gaussian noise $w_t \in \mathcal{N}(0, id)$

$$A = \begin{bmatrix} 1.01 & 0.01 & 0 \\ 0.01 & 1.01 & 0.01 \\ 0 & 0.01 & 1.01 \end{bmatrix}, \quad B = I$$

- policy gradient methods are more **robust** to noise versus sensitive one-shot method
- **empirically** observe tighter optimality gap $\sim O(SNR^{-2})$ than our certificate $O(SNR^{-1})$

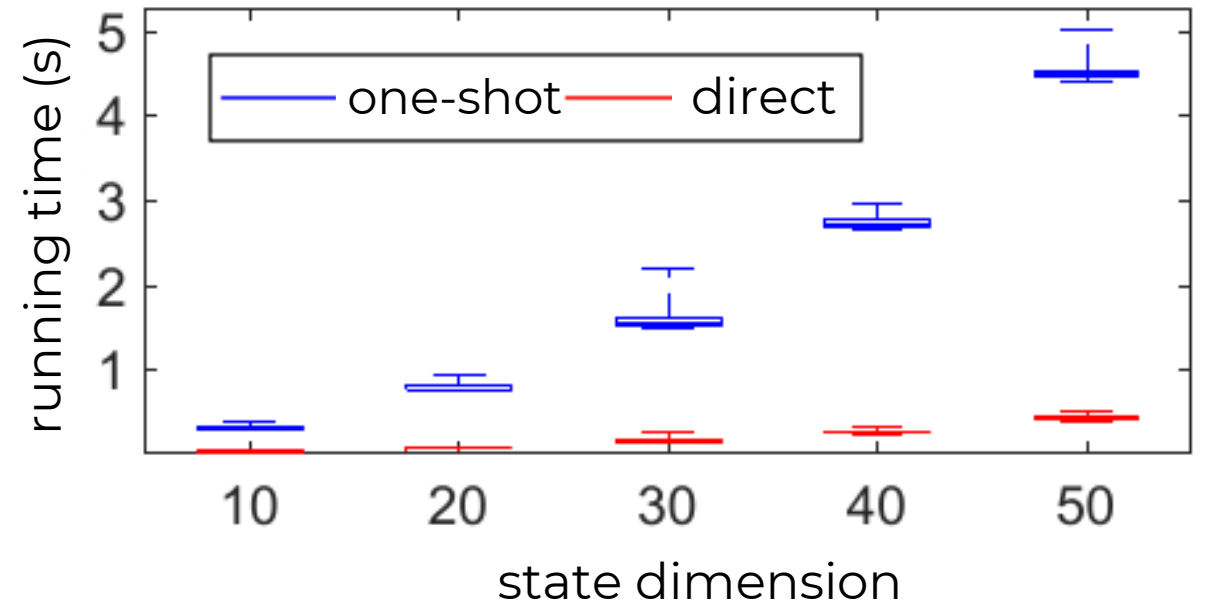
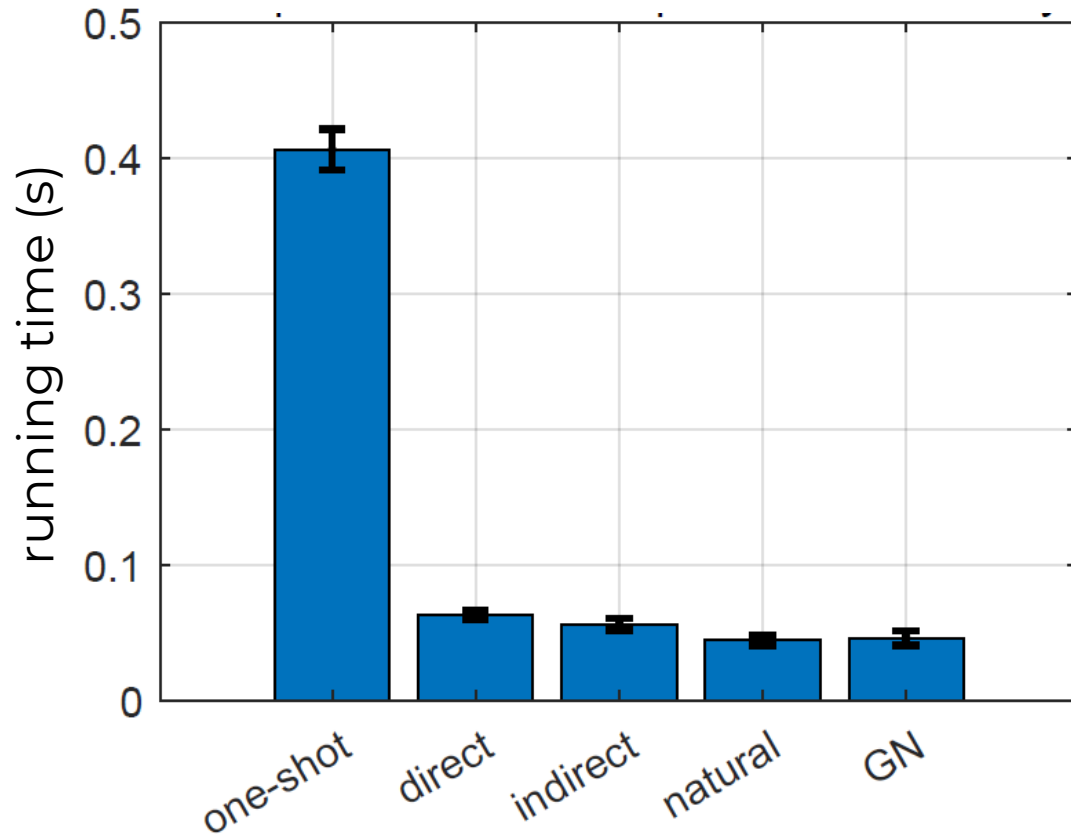


Numerics: mean $\pm$ std of closed-loop realized cost



→ all converge with a bias, but one-shot & Gauss-Newton less robust

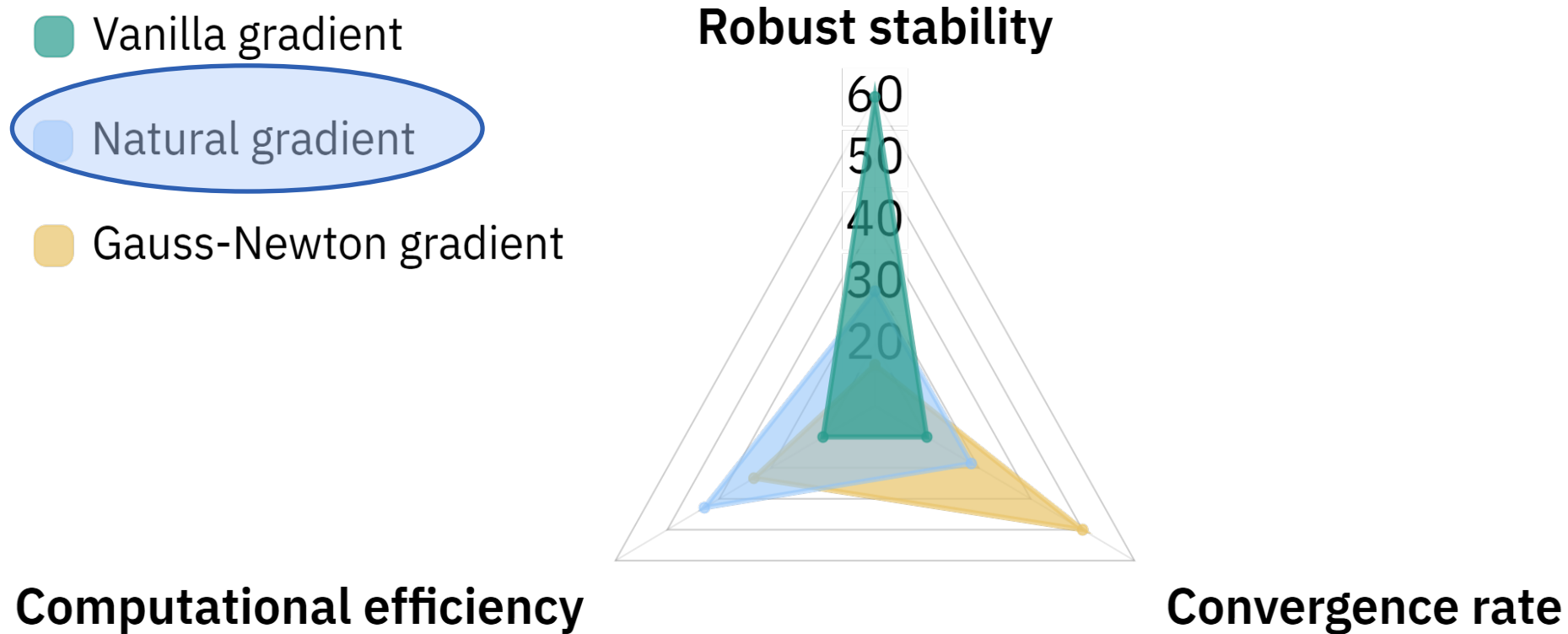
Numerics: computational efficiency



→ all policy gradient methods significantly outperform one-shot-based method in **computational effort**

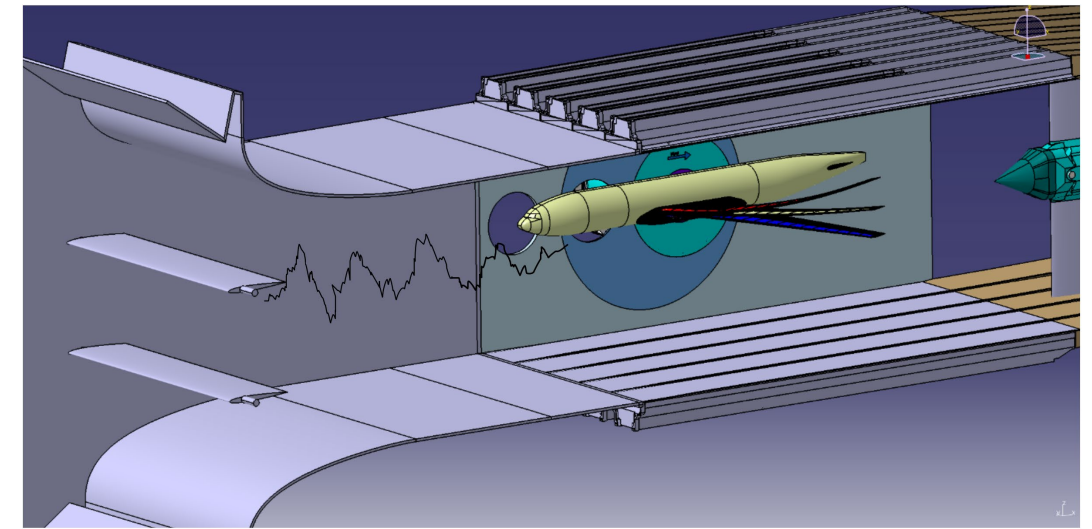
Comparison among different gradients

- Vanilla gradient
- Natural gradient
- Gauss-Newton gradient

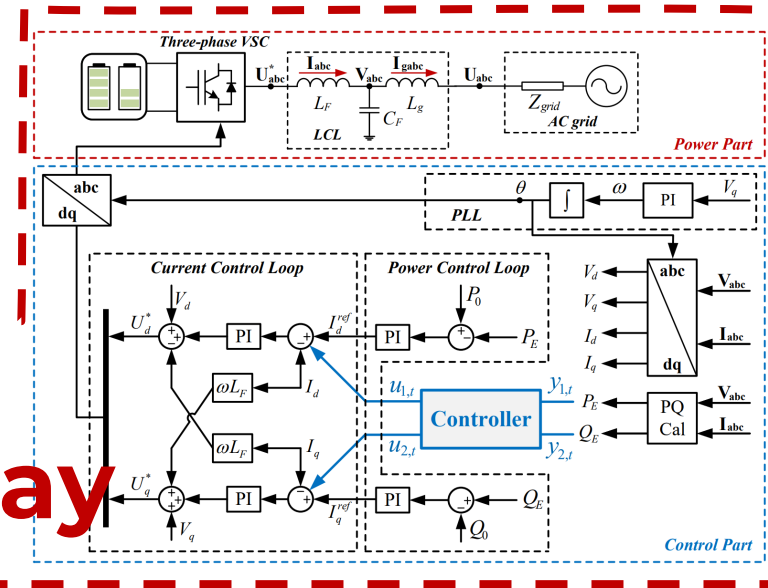


→ Best option: natural gradient (fast computing & convergence)+ regularization (promoting robustness)

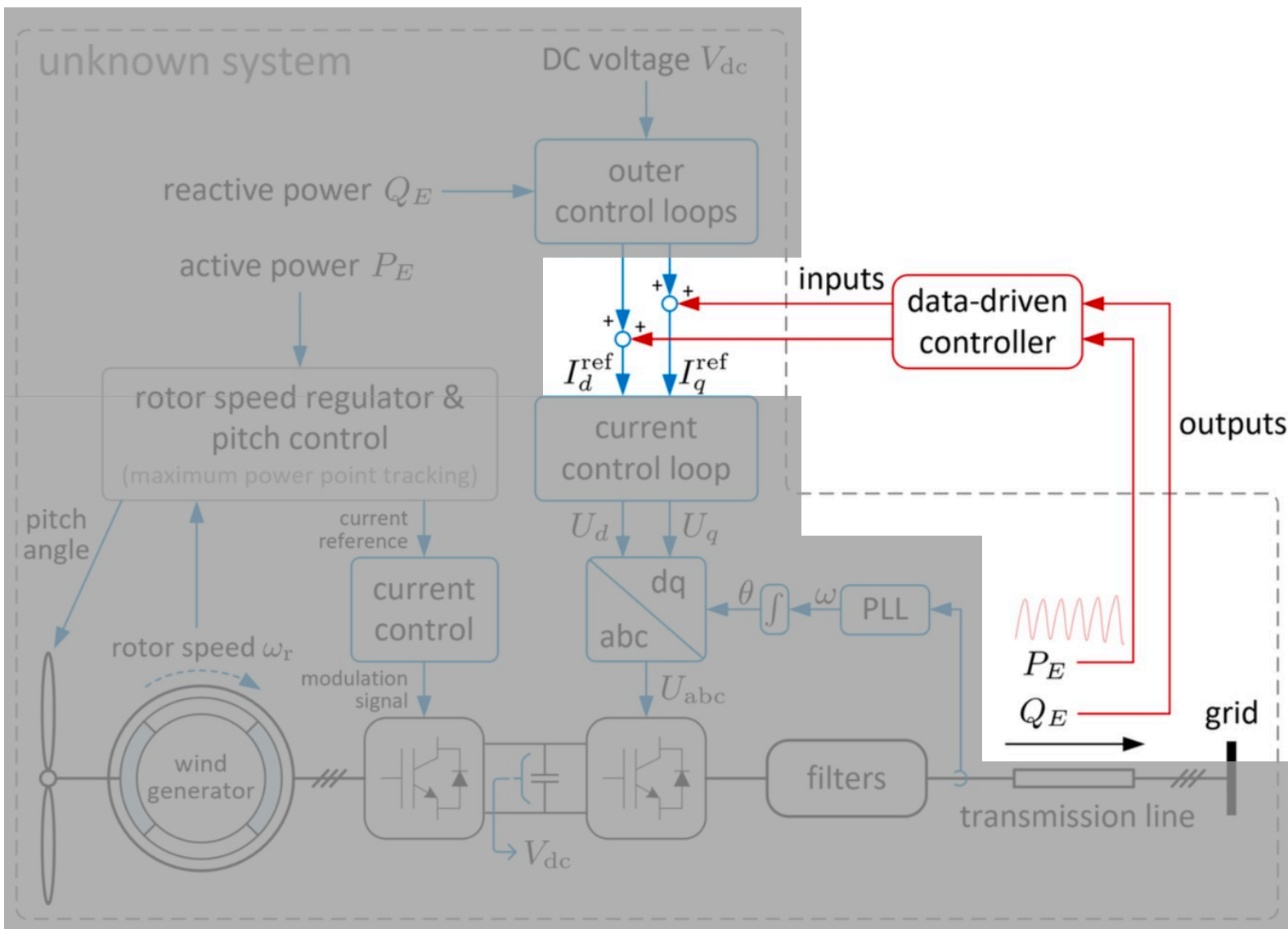
Implementations



today



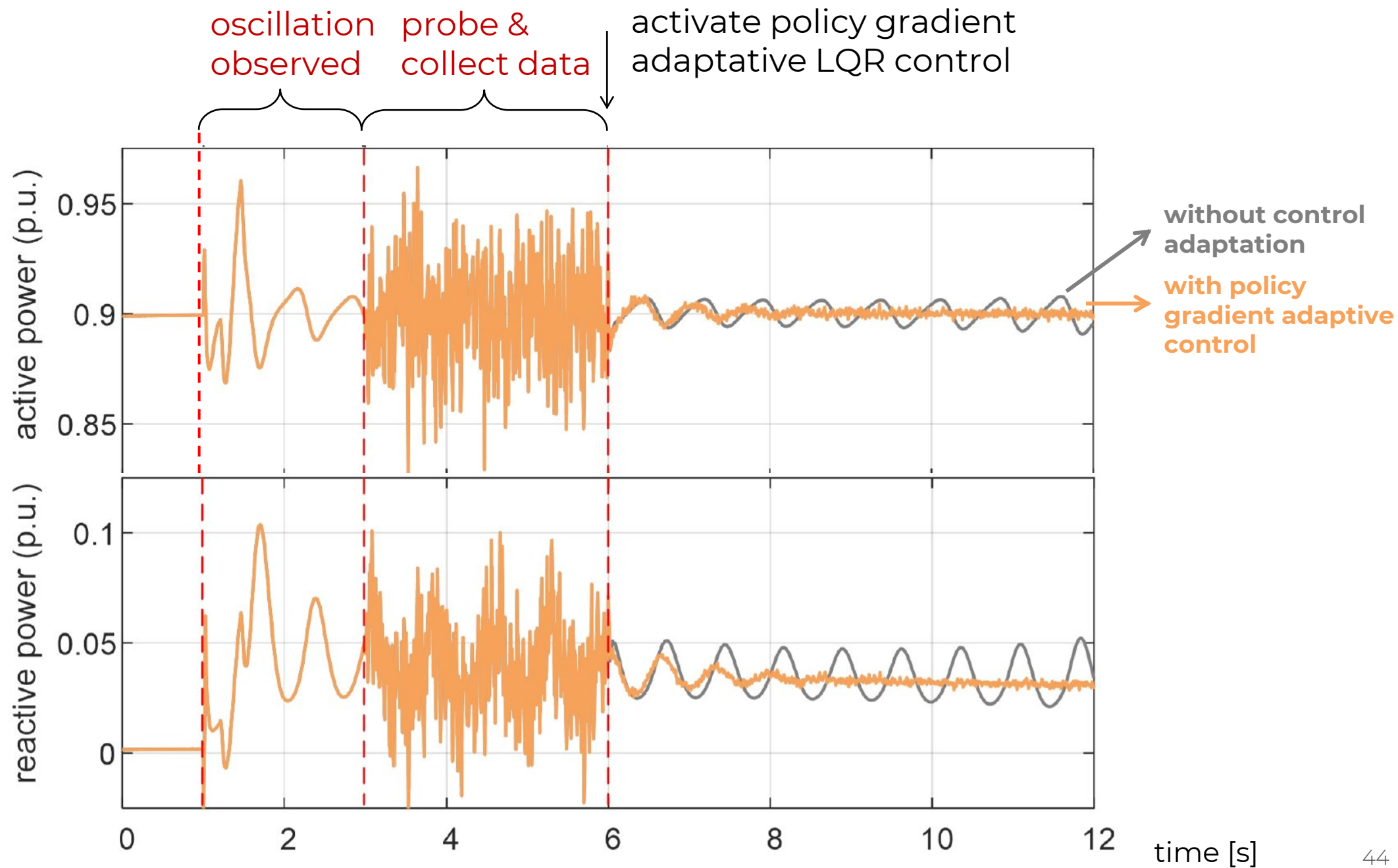
Power systems / electronics case study



synchronous generator & full-scale converter

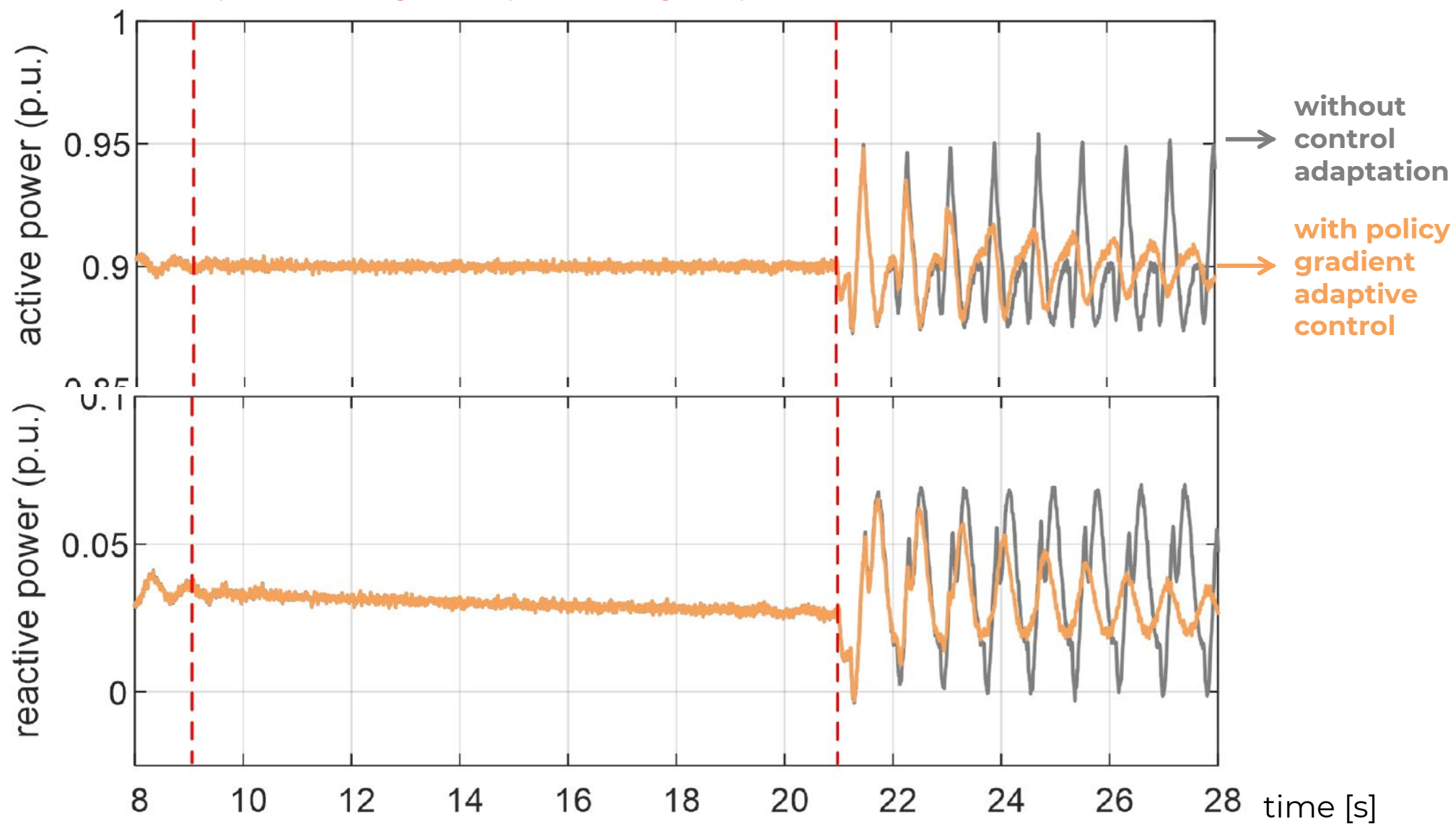
- wind turbine becomes **unstable** in weak grid → nonlinear oscillations
- converter, turbine, & grid are **black box** for commission engineer
- construct **state space realization** from time shifts (5ms sampling) of inputs & outputs

→ **direct policy gradient**



change of system parameters
(DC voltage setpoint & gain)

AC grid voltage
disturbance occurs



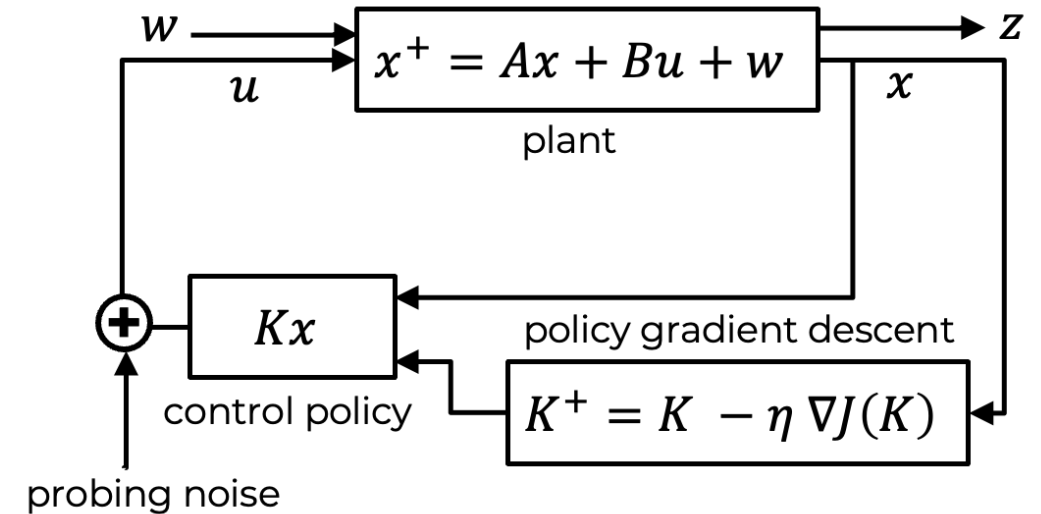
Conclusions

Summary

- policy gradient adaptive control
- various algorithmic pipelines
- closed-loop stability & optimality
- academic & real-world case studies

Future work

- **technicalities**: weaken assumptions & improve rates
- **extend problem setting**: output feedback, other objectives & complex system classes: stochastic, time-varying, & nonlinear
- **persistency of excitation**: how to best design probing noise



THANKS